Computer Networks

- A computer network is a system of interconnected devices that can communicate and share data or resources with each other.
- The devices in a network can be computers, servers, routers, switches, printers, scanners, cameras, etc.
- The devices are connected by physical or wireless media, such as cables, optical fibers, radio waves, or infrared signals.
- The network can be classified by its size, topology, architecture, or protocol.
- The size of a network can range from a personal area network (PAN) that connects a few devices within a few meters, to a wide area network (WAN) that spans across continents or even the globe.
- The topology of a network refers to the shape or layout of the connections between the devices. Some common topologies are bus, ring, star, tree, mesh, and hybrid.
- The architecture of a network defines how the devices are organized and how they communicate with each other. Some common architectures are peer-to-peer (P2P), client-server, and cloud computing.
- The protocol of a network is a set of rules or standards that govern the format, transmission, and exchange of data or messages between the devices. Some common protocols are TCP/IP, Ethernet, Wi-Fi, Bluetooth, and HTTP.
- The main functions of a network are to enable data communication, resource sharing, distributed processing, and network security.

Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

- A computer network is a collection of devices that can communicate with each other using a common set of rules or protocols.
- The main components of a computer network are hosts, links, switches, routers, and the Internet.
- Hosts are devices that run applications and generate or consume data, such as computers, smartphones, servers, etc.
- Links are physical media that connect hosts and switches, such as cables, wireless channels, etc.
- Switches are devices that forward data within a local area network (LAN), such as Ethernet switches, Wi-Fi access points, etc.
- Routers are devices that forward data across different networks, such as the Internet, using IP addresses.
- The Internet is a global network of networks that connects billions of hosts using the Internet Protocol (IP).
- A computer network can be classified by its size, topology, architecture, or performance.
- The size of a network refers to the number and distance of hosts and

links, such as personal area network (PAN), local area network (LAN), metropolitan area network (MAN), or wide area network (WAN).

- The topology of a network refers to the shape or layout of the links and switches, such as bus, ring, star, tree, or mesh.
- The architecture of a network refers to the design or organization of the protocols and layers, such as peer-to-peer (P2P), client-server, or hybrid.
- The performance of a network refers to the quality or efficiency of the data transmission, such as bandwidth, latency, jitter, or reliability.
- A computer network can be modeled using a layered approach, such as the Open Systems Interconnection (OSI) model or the Transmission Control Protocol/Internet Protocol (TCP/IP) model.
- The OSI model consists of seven layers: physical, data link, network, transport, session, presentation, and application.
- The TCP/IP model consists of four layers: network access, internet, transport, and application.
- The physical layer is the lowest layer of the OSI model and the network access layer of the TCP/IP model.
- The physical layer is responsible for transmitting bits over a link using electrical, optical, or radio signals.
- The physical layer defines the characteristics of the link, such as the type, speed, frequency, modulation, encoding, or multiplexing of the signals.
- The physical layer also defines the standards or specifications for the connectors, cables, and interfaces of the devices, such as RJ-45, USB, HDMI, etc.
- The physical layer can be divided into two sublayers: the physical medium dependent (PMD) sublayer and the physical medium independent (PMI) sublayer.
- The PMD sublayer deals with the actual transmission and reception of the signals over the medium, such as copper, fiber, or air.
- The PMI sublayer deals with the encoding and decoding of the signals, such as Manchester, NRZ, or QAM.

Introductory Concepts of Computer Networks

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- The devices are connected by physical or wireless media, such as cables, optical fibers, radio waves, etc.
- The network can be classified by its size, topology, architecture, or protocol.
- The size of a network refers to the number and distance of the devices in the network. Some common network sizes are:
  - Local Area Network (LAN): A network that covers a small geographic area, such as a home, office, or building.

  - Metropolitan Area Network (MAN): A network that covers a large city or metropolitan area, such as a campus or a city.
  - Wide Area Network (WAN): A network that covers a large geographic area, such as a country or a continent.
- The topology of a network refers to the shape or layout of the connections between the devices in the network. Some common network topologies are:
  - Bus: A network where all the devices are connected to a single cable or backbone.
  - Star: A network where all the devices are connected to a central device, such as a hub or a switch.
  - Ring: A network where all the devices are connected in a circular fashion, forming a closed loop.
  - Mesh: A network where each device is connected to every other device, forming a web of connections.
  - Tree: A network where the devices are connected in a hierarchical structure, forming a tree-like shape.
- The architecture of a network refers to the design or model of the network, such as how the devices are organized, how the data is transmitted, and how the network is managed. Some common network architectures are:
  - Peer-to-peer (P2P): A network where each device can act as both a client and a server, and can communicate directly with any other device in the network.
  - Client-server: A network where some devices act as servers, providing services or resources to other devices, and some devices act as clients, requesting services or resources from the servers.
  - Hybrid: A network that combines the features of both P2P and client-server architectures, such as having some centralized servers and some distributed peers.
- The protocol of a network refers to the set of rules or standards that govern the communication and data exchange between the devices in the network. Some common network protocols are:
  - Transmission Control Protocol/Internet Protocol (TCP/IP): A suite of protocols that defines how data is divided, addressed, routed, and delivered over the Internet or other networks.
  - Hypertext Transfer Protocol (HTTP): A protocol that defines how web browsers and web servers communicate and exchange web pages and other resources over the Internet.
  - File Transfer Protocol (FTP): A protocol that defines how files are transferred between devices over a network.
  - Simple Mail Transfer Protocol (SMTP): A protocol that defines how email messages are sent and received over a network.

Goals and applications of networks and protocols

- A network is a collection of devices that can communicate with each other over a shared medium, such as a cable, a wireless link, or the internet.

- A protocol is a set of rules that defines how devices exchange information over a network, such as how to format, transmit, and receive data packets, how to handle errors, and how to ensure reliability and security.
- The main goals of networks and protocols are:
  - **Resource sharing**: Networks enable devices to share hardware and software resources, such as printers, scanners, files, databases, applications, etc. This reduces the cost and increases the efficiency of using these resources. For example, a group of office workers can use a common printer connected to a network instead of having individual printers for each workstation.
  - **High reliability**: Networks provide multiple sources of supply and backup for data and services, which increases the reliability and availability of the network. For example, if one server fails, another server can take over its tasks and prevent data loss or service interruption.
  - **Greater flexibility**: Networks allow devices to connect and communicate with each other regardless of their physical location, type, or configuration. This enables users to access data and services from anywhere, anytime, and from any device. For example, a user can access their email account from their laptop, smartphone, or tablet using a network.
  - **Increased productivity**: Networks facilitate the exchange of information and collaboration among users, which improves the productivity and performance of individuals and organizations. For example, a team of workers can use a network to share documents, chat, video conference, and work on a project together.
  - **Enhanced communication**: Networks enable the transmission of various types of data, such as text, audio, video, images, etc., over different distances and speeds, using different media and technologies. This enhances the communication and interaction among users and devices. For example, a user can use a network to send an email, make a phone call, watch a video, or browse a website.
- Some common applications of networks and protocols are:
  - **The internet**: The internet is a global network of networks that connects billions of devices and users using various protocols, such as TCP/IP, HTTP, SMTP, DNS, etc. The internet enables the access and delivery of various services and resources, such as web pages, email, online shopping, social media, etc. The internet is the largest and most widely used network in the world.
  - **Local area network (LAN)**: A LAN is a network that connects devices within a small geographic area, such as a home, office, or school, using protocols such as Ethernet, Wi-Fi, Bluetooth, etc. A LAN enables the sharing of resources and data among devices within

the same network. A LAN is usually owned and managed by a single entity, such as a person, a company, or an institution.

  - **Wide area network (WAN)**: A WAN is a network that connects devices across a large geographic area, such as a city, a country, or the world, using protocols such as IP, MPLS, ATM, etc. A WAN enables the communication and data transfer among devices that are not in the same network. A WAN is usually composed of multiple LANs and other networks, and is owned and managed by multiple entities, such as internet service providers, telecommunication companies, or governments.
  - **Wireless network**: A wireless network is a network that uses radio waves, infrared, or other electromagnetic signals to transmit data, instead of cables or wires. Wireless networks enable the mobility and portability of devices and users, as well as the connection of devices that are difficult or impractical to wire. Wireless networks can be classified into different types, such as cellular networks, Wi-Fi networks, Bluetooth networks, satellite networks, etc., depending on the technology, frequency, range, and purpose of the network.
  - **Peer-to-peer network**: A peer-to-peer network is a network that does not have a central server or authority that controls the communication and data exchange among devices. Instead, each device acts as both a client and a server, and can directly communicate and share resources with other devices in the network. Peer-to-peer networks are decentralized, distributed, and self-organizing, and can be used for various applications, such as file sharing, streaming, gaming, etc.

Categories of networks in computer networks

- A computer network is a collection of devices that are connected by communication channels to share data and resources.
- There are different ways to categorize computer networks based on their size, scope, topology, architecture, and purpose.
- Some of the common categories of networks are:
  - **Local Area Network (LAN)**: A LAN is a network that connects devices within a small geographic area, such as a home, office, or building. LANs typically use wired or wireless technologies, such as Ethernet, Wi-Fi, or Bluetooth, to transmit data at high speeds. LANs are often used to share files, printers, scanners, and internet access among devices.
  - **Wide Area Network (WAN)**: A WAN is a network that connects devices across a large geographic area, such as a city, country, or continent. WANs typically use leased lines, satellite links, or cellular networks to transmit data at lower speeds than LANs. WANs are often used to connect LANs or other networks over long distances,

such as the internet.

  - **Metropolitan Area Network (MAN)**: A MAN is a network that connects devices within a metropolitan area, such as a city or a campus. MANs are usually larger than LANs but smaller than WANs. MANs typically use fiber-optic cables, microwave links, or wireless technologies to transmit data at high speeds. MANs are often used to provide internet access, cable TV, or telephony services to a large population.
  - **Personal Area Network (PAN)**: A PAN is a network that connects devices within a personal range, such as a few meters or centimeters. PANs typically use wireless technologies, such as Bluetooth, infrared, or NFC, to transmit data at low speeds. PANs are often used to connect personal devices, such as smartphones, laptops, tablets, or wearable devices.
  - **Storage Area Network (SAN)**: A SAN is a network that connects storage devices, such as hard disks, solid state drives, or tape drives, to servers or other devices. SANs typically use fiber-optic cables, SCSI, or iSCSI protocols to transmit data at high speeds. SANs are often used to provide centralized, high-performance, and reliable storage for data-intensive applications, such as databases, video streaming, or backup.
  - **Campus Area Network (CAN)**: A CAN is a network that connects devices within a campus, such as a university, school, or hospital. CANs are usually composed of multiple LANs that are interconnected by routers or switches. CANs typically use wired or wireless technologies, such as Ethernet, Wi-Fi, or fiber-optic cables, to transmit data at high speeds. CANs are often used to provide internet access, email, or e-learning services to students, staff, or visitors.

Organization of the Internet

- The Internet is a global network of interconnected computers and devices that communicate using standardized protocols and procedures.
- The Internet is composed of many smaller networks, called internets, that may or may not be connected to the global Internet.
- The Internet is organized at four levels: hardware, access, navigation, and community.
  - Hardware: The physical devices and infrastructure that enable data transmission and processing, such as routers, switches, cables, servers, etc.
  - Access: The providers and intermediaries that offer Internet connectivity and services to users, such as Internet service providers (ISPs), Internet exchange points (IXPs), content delivery networks (CDNs), etc.
  - Navigation: The tools and platforms that help users find and access information and resources on the Internet, such as domain name

system (DNS), search engines, web browsers, etc.
  - Community: The sites and groups that create and share content and rules on the Internet, such as websites, blogs, social media, online forums, etc.
- The Internet is governed by a variety of actors and institutions that collaborate and coordinate through voluntary and consensus-based processes, such as Internet standards, policies, and best practices.
  - Some of the key actors and institutions in the Internet ecosystem are:
    * Internet Engineering Task Force (IETF): A technical body that develops and publishes Internet standards and protocols.
    * Internet Corporation for Assigned Names and Numbers (ICANN): A non-profit organization that manages the global domain name system and IP address allocation.
    * Internet Society (ISOC): A non-profit organization that advocates for the open, secure, and trustworthy Internet for everyone.
    * World Wide Web Consortium (W3C): A technical body that develops and publishes web standards and guidelines.
    * Regional Internet Registries (RIRs): Five regional organizations that manage the distribution and registration of IP addresses and autonomous system numbers in their respective regions.
    * Internet Governance Forum (IGF): A multi-stakeholder platform that facilitates dialogue and cooperation on Internet governance issues.

ISP

- ISP stands for Internet Service Provider .
- An ISP is a company that provides access to the internet to individuals and organizations .
- An ISP can provide internet access through various means, such as dial-up, DSL, cable, wireless and fiber-optic connections .
- An ISP may also offer other services, such as email, domain registration, web hosting, browser services, and security software .
- An ISP can host websites for businesses and can also build the websites themselves.
- An ISP may charge a monthly fee for its services, depending on the type and speed of internet connection, the amount of data usage, and the features included .
- An ISP may have different plans and packages for different customers, such as residential, commercial, or institutional .
- An ISP may compete with other ISPs in the same area or region, based on the quality, reliability, and affordability of their services .
- An ISP may be regulated by the government or other authorities, depending on the country or jurisdiction, to ensure fair competition, consumer protection, and net neutrality .

Network structure with reference to Computer Networks

- A computer network is a structure that makes available to a data processing user at one place some data processing function or service performed at another place.
- A computer network consists of two or more devices that can communicate with each other and share data, files, resources, or services.
- A computer network can be classified by its size, topology, architecture, and protocols.
- The size of a network refers to the geographical area it covers and the number of devices it connects. Some common network sizes are:
  - LAN (local area network): A LAN connects computers over a relatively short distance, such as within a building or a campus. A LAN typically uses wired media, such as Ethernet cables, to transfer data at high speeds.
  - WLAN (wireless local area network): A WLAN is just like a LAN but connections between devices on the network are wireless, such as Wi-Fi or Bluetooth. A WLAN offers more mobility and flexibility than a LAN, but may have lower security and reliability.
  - MAN (metropolitan area network): A MAN connects computers over a larger area, such as a city or a region. A MAN may use a combination of wired and wireless media, such as fiber-optic cables and microwave links, to provide high-speed data transmission.
  - WAN (wide area network): A WAN connects computers over a very large area, such as a country or a continent. A WAN may use various media and technologies, such as satellite, telephone lines, and cellular networks, to provide long-distance data communication. The Internet is the largest and most well-known example of a WAN.
- The topology of a network refers to the physical or logical arrangement of the devices and the links that connect them. Some common network topologies are:
  - Bus: A bus network has a single cable that connects all the devices on the network. A bus network is simple and cheap to implement, but may have low performance and reliability if the cable is damaged or overloaded.
  - Star: A star network has a central device, such as a switch or a hub, that connects all the other devices on the network. A star network is easy to expand and troubleshoot, but may have high cost and dependency on the central device.
  - Ring: A ring network has a circular arrangement of devices that are connected by a single cable. A ring network has high performance and reliability, as data can travel in both directions and there is no collision, but may have difficulty in adding or removing devices.
  - Mesh: A mesh network has multiple connections between the devices on the network, creating a web-like structure. A mesh network has high fault tolerance and scalability, as data can take multiple paths

to reach the destination, but may have high complexity and cost.
  - Tree: A tree network has a hierarchical structure of devices that are connected by branches of cables. A tree network is a combination of bus and star topologies, and has the advantages and disadvantages of both.
- The architecture of a network refers to the way network devices and services are structured and organized to serve the connectivity needs of client devices. Some common network architectures are:
  - Client-server: A client-server network has one or more servers that provide services, such as file storage, email, or web hosting, to the client devices that request them. A client-server network has high efficiency and security, as the servers can manage the resources and the access control, but may have high cost and maintenance.
  - Peer-to-peer: A peer-to-peer network has no servers, and each device on the network can act as both a client and a server, sharing its own resources and services with other devices. A peer-to-peer network has low cost and easy setup, as no central administration is required, but may have low security and reliability, as the devices may not be always available or trustworthy.
- The protocols of a network refer to the rules and standards that govern the communication and data exchange between the devices on the network. Some common network protocols are:
  - TCP/IP: TCP/IP is the predominant model for today's Internet structure and presents this standard layer configuration for communication links:
    * Network access layer: Defines how the data gets physically transferred.
    * Internet layer: Packages the data into understandable packets so it can be sent and received.
    * Transport layer: Allows the network devices to maintain conversations.
    * Application layer: Establishes how high-level applications access the network for purposes of data transfer.
  - Ethernet: Ethernet is a protocol that defines how data is transmitted over a LAN using a bus or a star topology. Ethernet uses a technique called CS

Network architecture with reference to Computer Networks

- Network architecture is the design of a computer network .
- It is a framework for the specification of a network's physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used .
- Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols .

- Network architecture can be classified based on the network's size and purpose, such as LAN (local area network), WLAN (wireless local area network), WAN (wide area network), MAN (metropolitan area network), CAN (campus area network), SAN (storage area network), PAN (personal area network), etc .
- Network architecture can also be categorized based on the network's design approach, such as peer-to-peer, client-server, cloud computing, etc .
- Network architecture can affect the network's performance, security, scalability, reliability, and manageability .
- Network architecture can be designed using various models and standards, such as OSI (open systems interconnection), TCP/IP (transmission control protocol/internet protocol), etc .

Layering Principles with Reference to Network Architecture in Computer Networks

- Network architecture is the design and structure of a communication system that consists of various components and protocols that work together to enable data transmission and exchange.

- Layering is a mechanism that divides the complex network architecture into smaller and manageable parts, called layers, that perform specific functions and interact with each other through well-defined interfaces.

- Layering principles are the guidelines or rules that are used to design and implement the network architecture using layers. Some of the common layering principles are:

  - A layer should be created where a different abstraction is needed. Abstraction is the process of hiding the details of how a function is performed and exposing only the essential features to the higher layer. For example, the physical layer abstracts the details of how bits are transmitted over a medium and provides a bit stream service to the data link layer.
  - Each layer should perform a well-defined function that is distinct from the functions of other layers. For example, the network layer is responsible for routing packets across different networks, while the transport layer is responsible for ensuring reliable and ordered delivery of data between end hosts.
  - The function of each layer should be chosen with an eye toward defining internationally standardized protocols that can be used by different vendors and systems. For example, the Internet Protocol (IP) is a widely adopted network layer protocol that enables interoperability among different networks and devices.
  - The layer boundaries should be chosen to minimize the information flow across the interfaces. This means that each layer should only exchange the necessary information with the adjacent layers and avoid

passing redundant or irrelevant data. For example, the data link layer only adds a header and a trailer to the network layer packet and does not modify or inspect its contents.
  - The number of layers should be large enough to separate the different functions, but small enough to keep the architecture simple and efficient. There is no definitive answer to how many layers are optimal, but a common model that is widely used is the Open Systems Interconnection (OSI) model, which has seven layers: physical, data link, network, transport, session, presentation, and application.

Services in Networks Architecture in Computer Networks

- Services in networks architecture are the set of primitive operations that each layer provides to the layer above it .
- Services are typically implemented by protocols, which are rules and formats for exchanging data between network entities.
- Services can be classified into two types: connection-oriented and connectionless.
  - Connection-oriented services require the establishment of a logical connection between the sender and the receiver before data can be transferred. The connection provides reliable, ordered, and error-free delivery of data. An example of a connection-oriented service is TCP.
  - Connectionless services do not require a connection and allow data to be sent without prior arrangement. The data may be lost, duplicated, or delivered out of order. An example of a connectionless service is UDP.
- Services can also be classified into two categories: peer-to-peer and client-server.
  - Peer-to-peer services allow network devices to communicate directly with each other without the need for a central server. Each device can act as both a client and a server. An example of a peer-to-peer service is BitTorrent.
  - Client-server services involve a central server that provides services to multiple clients. The server typically has more resources and capabilities than the clients. An example of a client-server service is HTTP.
- Some examples of services in networks architecture are:
  - DHCP: Dynamic Host Configuration Protocol, which assigns IP addresses and other network parameters to devices.
  - DNS: Domain Name System, which translates domain names to IP addresses and vice versa.
  - FTP: File Transfer Protocol, which allows file transfer between devices.
  - SMTP: Simple Mail Transfer Protocol, which enables email delivery.
  - SSH: Secure Shell, which provides secure remote access to devices.
  - VPN: Virtual Private Network, which creates a secure tunnel between

devices over a public network.

Protocols and Standards in Networks Architecture in Computer Networks

- A **protocol** is a set of rules or algorithms that define how two or more devices can communicate across a network. Protocols specify the formats, procedures, and rules for data exchange, error handling, synchronization, and security.
- A **standard** is a formal document that establishes uniform engineering or technical criteria, methods, processes, and practices for a particular domain. Standards ensure interoperability, compatibility, and reliability among different devices, vendors, and applications.
- Network architecture is the design and structure of a network, including its components, connections, protocols, and standards. Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols.
- There are two main types of network architecture: **peer-to-peer (P2P)** and **client/server**. In P2P architecture, two or more computers are connected as "peers" and share resources and data without a central authority. In client/server architecture, one or more computers act as "servers" that provide services and resources to other computers, called "clients", that request them.
- There are different protocols and standards defined at each layer of the **OSI model** and the **TCP/IP model**, which are two common frameworks for describing the functions and interactions of network layers. Some examples of protocols and standards are:
  - **TCP**: Transmission Control Protocol, a reliable and connection-oriented protocol that ensures data delivery and error recovery at the transport layer.
  - **IP**: Internet Protocol, a connectionless and best-effort protocol that provides addressing and routing functions at the network layer.
  - **UDP**: User Datagram Protocol, a connectionless and unreliable protocol that provides fast and simple data transmission at the transport layer.
  - **ARP**: Address Resolution Protocol, a protocol that maps network layer addresses (IP) to data link layer addresses (MAC) at the network layer.
  - **DHCP**: Dynamic Host Configuration Protocol, a protocol that assigns IP addresses and other network configuration parameters to devices dynamically at the application layer.
  - **FTP**: File Transfer Protocol, a protocol that enables file transfer between hosts at the application layer.
  - **SMTP**: Simple Mail Transfer Protocol, a protocol that enables email delivery between hosts at the application layer.
  - **DNS**: Domain Name System, a protocol that resolves domain names

to IP addresses at the application layer.
  - **IEEE**: Institute of Electrical and Electronics Engineers, a standard organization that develops and publishes standards for various fields, including computer networks.
  - **IETF**: Internet Engineering Task Force, a standard organization that develops and publishes standards for the Internet, such as TCP/IP.
  - **ISO**: International Organization for Standardization, a standard organization that develops and publishes standards for various fields, including the OSI model.

The OSI reference model in Computer Networks

The OSI reference model is a conceptual framework that defines how different devices and applications communicate over a network. It was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.

The OSI reference model consists of seven layers, each of which performs a specific function in the network communication process. The layers are:

- Layer 1: Physical. This layer defines the physical characteristics of the transmission medium, such as the voltage, frequency, modulation, and connectors. It also deals with the transmission and reception of raw bits over the medium.
- Layer 2: Data Link. This layer provides reliable and error-free data transfer between adjacent nodes on the same network segment. It also handles the framing, addressing, and flow control of data packets. It can be divided into two sublayers: Logical Link Control (LLC) and Media Access Control (MAC).
- Layer 3: Network. This layer provides logical addressing and routing of data packets across different network segments. It also handles the fragmentation and reassembly of packets, as well as the congestion control and quality of service (QoS) of the network.
- Layer 4: Transport. This layer provides end-to-end data delivery between applications on different hosts. It also handles the segmentation and reassembly of data streams, as well as the error detection and correction, and flow control of the data. It can support different transport protocols, such as TCP and UDP.
- Layer 5: Session. This layer establishes, maintains, and terminates sessions between applications on different hosts. It also handles the synchronization, authentication, and authorization of the communication.
- Layer 6: Presentation. This layer provides data representation and encryption for the applications. It also handles the compression, decompression, and translation of data formats, such as ASCII, EBCDIC, JPEG, and MPEG.
- Layer 7: Application. This layer provides the interface and services for the applications to access the network. It also handles the high-level functions,

such as email, file transfer, web browsing, and remote access.

The OSI reference model is useful for understanding how different network components and protocols work together, as well as for designing and troubleshooting network systems. However, it is not a strict standard that must be followed by all network implementations. Some protocols, such as IP, do not fit neatly into the OSI model, and some layers, such as the session and presentation layers, are often combined or omitted in practice. Therefore, the OSI reference model should be seen as a general guideline, rather than a rigid rule, for network communication.

TCP/IP protocol suite in in Computer Networks

- TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet or other computer networks .
- TCP/IP is based on a four-layer model that divides the communication functions into four abstraction levels: application, transport, internet, and link .
- The application layer provides the interface for the user applications to access the network services, such as web browsing, email, file transfer, etc. Some of the common protocols in this layer are HTTP, SMTP, FTP, DNS, etc .
- The transport layer provides reliable or unreliable data delivery services between the end hosts, such as error detection, flow control, congestion control, etc. The main protocols in this layer are TCP and UDP .
- The internet layer provides the routing and addressing functions for the data packets across different networks, such as IP, ICMP, ARP, etc. The internet layer is responsible for the logical addressing scheme of the network, such as IPv4 and IPv6 .
- The link layer provides the physical and data link functions for the data transmission over a single network segment, such as Ethernet, Wi-Fi, etc. The link layer is responsible for the physical addressing scheme of the network, such as MAC addresses .
- TCP/IP is the most widely used protocol suite in the world and is the basis of the internet. It is also used as a communications protocol in private networks, such as intranets and extranets .

Network devices in Computer Networks

Network devices are physical devices that enable communication and interaction between hardware on a computer network. Each networking device operates in a distinct computer network segment and performs distinct functions. A network may require hundreds or thousands of different network devices to maintain and build out various LAN and WAN.

Some of the common types of network devices are:

- **Repeater**: A repeater is a device that operates at the physical layer and regenerates the signal over the same network. It can extend the transmission distance of a network by amplifying the weak signals.
- **Hub**: A hub is a device that operates at the physical layer and connects multiple wires coming from different branches. It broadcasts the data to all the connected devices, regardless of the destination address. It is also known as a multiport repeater.
- **Bridge**: A bridge is a device that operates at the data link layer and connects two or more network segments. It filters the data based on the MAC addresses and forwards only the relevant data to the destination segment. It can also reduce network congestion by dividing a large network into smaller segments.
- **Switch**: A switch is a device that operates at the data link layer or the network layer and connects multiple devices on the same network. It stores the MAC addresses of the connected devices in a table and forwards the data only to the intended device, based on the destination address. It can improve the network efficiency and security by reducing collisions and broadcast traffic .
- **Router**: A router is a device that operates at the network layer and connects two or more different networks. It routes the data packets based on their IP addresses and the best available path. It can also perform network address translation (NAT), firewall, and encryption functions .
- **Gateway**: A gateway is a device that operates at the application layer and connects two or more networks that use different protocols. It converts the data from one format to another, based on the destination network's requirements. It can also perform protocol conversion, data encryption, and authentication functions.
- **Brouter**: A brouter is a device that combines the functions of a bridge and a router. It operates at both the data link layer and the network layer and can filter and route the data based on the MAC addresses or the IP addresses. It can also act as a switch for the same network and as a router for different networks.
- **NIC**: A network interface card (NIC) is a device that operates at the physical layer and the data link layer and connects a computer or a device to a network. It provides a unique MAC address for the device and enables data transmission and reception over the network. It can also perform error detection and correction functions.

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Network Components in Computer Networks

Network components are the devices and software that enable a computer system to communicate with other systems or networks. They are essential for data transmission, routing, switching, and security. Some of the major network components are:

- **Network Interface Card (NIC)**: A NIC is a hardware device that connects a computer to a network. It provides a physical interface for sending and receiving data over a network media, such as a cable or a wireless signal. A NIC has a unique address, called a MAC address, that identifies the computer on the network .
- **Hub**: A hub is a device that splits a network connection into multiple ports, allowing multiple computers to share the same network segment. A hub operates at the physical layer of the network, and does not perform any filtering or processing of the data. A hub broadcasts all the data it receives to all the ports, which can cause collisions and congestion on the network .
- **Switch**: A switch is a device that connects multiple devices on a network and forwards data packets to the appropriate destination. A switch operates at the data link layer of the network, and uses the MAC addresses of the devices to determine where to send the data. A switch can reduce collisions and improve network performance by creating separate collision domains for each port .
- **Cables and connectors**: Cables and connectors are the physical media that carry the data signals between the network devices. There are different types of cables and connectors, depending on the speed, distance, and environment of the network. Some common examples are twisted pair cables, coaxial cables, fiber optic cables, RJ-45 connectors, BNC connectors, and SC connectors .
- **Router**: A router is a device that connects multiple networks and forwards data packets between them. A router operates at the network layer of the network, and uses the IP addresses of the devices to determine the best path for the data. A router can also perform network address translation (NAT), firewall, and quality of service (QoS) functions .
- **Modem**: A modem is a device that modulates and demodulates analog signals into digital signals, and vice versa. A modem allows a computer to communicate over a telephone line, a cable network, or a wireless network. A modem can also provide authentication and encryption functions .
- **Server**: A server is a computer that provides services or resources to other computers on a network. A server can perform various functions, such as file sharing, web hosting, email, database, printing, and backup. A server can be dedicated to a specific task, or can be multipurpose .
- **Client**: A client is a computer that requests services or resources from a server on a network. A client can run various applications, such as web browsers, email clients, word processors, and games. A client can be a desktop, a laptop, a tablet, a smartphone, or a smart device .
- **Firewall**: A firewall is a device or software that monitors and controls

the incoming and outgoing network traffic. A firewall can block or allow traffic based on predefined rules, such as the source and destination IP addresses, the port numbers, and the protocols. A firewall can provide network security and prevent unauthorized access .

- **Access point**: An access point is a device that allows wireless devices to connect to a wired network. An access point acts as a bridge between the wireless and wired networks, and can extend the range and coverage of the network. An access point can also provide encryption and authentication functions .

Physical Layer in Computer Networks

- The physical layer is the lowest layer of the OSI model, which deals with the transmission and reception of raw data bits over a physical medium.
- The physical layer defines the characteristics of the physical medium, such as the type of cable, connectors, voltage levels, modulation schemes, etc.
- The physical layer also defines the procedures for establishing, maintaining, and terminating a physical connection between two or more devices.
- The physical layer is responsible for converting the data bits into electrical, optical, or radio signals, and vice versa, depending on the medium used.
- The physical layer provides services such as bit synchronization, bit rate control, error detection, and multiplexing/demultiplexing of multiple channels over a single medium.
- The physical layer does not perform any error correction, encryption, compression, or routing functions, as these are handled by the higher layers of the OSI model.
- Some of the common standards and protocols used at the physical layer are Ethernet, Wi-Fi, Bluetooth, USB, RS-232, etc.

Network topology design in Computer Networks

- Network topology is the arrangement of the elements (nodes, links, etc.) of a communication network.
- Network topology can be used to define or describe the arrangement, logical or physical, of the different elements of a network.
- Network topology design is the process of selecting a network topology that meets the requirements and constraints of a given network scenario, such as performance, reliability, scalability, security, cost, etc.
- Network topology design can be influenced by various factors, such as:
  - The size and scope of the network (e.g., local area network, wide area network, etc.)
  - The type and number of devices and users in the network (e.g., computers, routers, switches, servers, etc.)
  - The bandwidth and latency requirements of the network applications and services (e.g., voice, video, data, etc.)
  - The availability and reliability of the network links and nodes (e.g.,

wired, wireless, optical, etc.)
  - The budget and resources available for the network deployment and maintenance (e.g., hardware, software, personnel, etc.)
  - The security and privacy policies and regulations of the network (e.g., encryption, authentication, access control, etc.)
- Network topology design can be classified into two main categories: physical topology and logical topology.
  - Physical topology refers to the actual layout of the network devices and cables, and how they are physically connected to each other.
  - Logical topology refers to the way the network devices communicate with each other, and how the data flows through the network, regardless of the physical layout.
- Network topology design can involve various types of topologies, such as:
  - Bus topology: A network topology in which all devices are connected to a single cable or backbone, and share the same communication channel.
  - Star topology: A network topology in which all devices are connected to a central device, such as a hub or a switch, and communicate through it.
  - Ring topology: A network topology in which all devices are connected in a closed loop, and data travels in one direction around the ring.
  - Mesh topology: A network topology in which every device is connected to every other device, and there are multiple paths for data transmission.
  - Tree topology: A network topology in which devices are connected in a hierarchical structure, and there is a root node that connects to other nodes, which in turn connect to other nodes, and so on.
  - Hybrid topology: A network topology that combines two or more of the above topologies, and can have different levels of complexity and flexibility.

Types of connections in Computer Networks

- A connection in a computer network is a link between two or more devices that enables them to communicate and exchange data.
- There are two main types of connections in computer networks: **point-to-point** and **point-to-multipoint**.
- A point-to-point connection is a direct link between two devices, such as a cable, a wireless link, or a fiber optic link. A point-to-point connection can be either **simplex**, **half-duplex**, or **full-duplex**.
  - A simplex connection allows data to flow in one direction only, such as a microphone to a speaker.
  - A half-duplex connection allows data to flow in both directions, but not at the same time, such as a walkie-talkie.
  - A full-duplex connection allows data to flow in both directions simultaneously, such as a telephone.

- A point-to-multipoint connection is a link between one device and multiple devices, such as a broadcast, a multicast, or a unicast. A point-to-multipoint connection can be either **one-to-many** or **many-to-many**.
  - A one-to-many connection allows data to flow from one device to multiple devices, such as a radio station to its listeners.
  - A many-to-many connection allows data to flow from multiple devices to multiple devices, such as a video conference.

Transmission media in Computer Networks

- Transmission media is the physical medium that carries data from one device to another in a computer network .
- Transmission media can be classified into two types: guided media and unguided media.
- Guided media, also known as wired or bounded media, uses cables or wires to transmit data signals. Examples of guided media are twisted pair, coaxial cable, and optical fiber.
- Unguided media, also known as wireless or unbounded media, uses electromagnetic waves or signals to transmit data through air, space, or water. Examples of unguided media are radio waves, microwaves, infrared waves, and visible light.
- The choice of transmission media depends on several factors, such as bandwidth, cost, distance, reliability, security, and interference .
- Transmission media is an essential component of network hardware, along with user devices, routers, servers, and gateways.
- Transmission media is also influenced by the network architecture, which is the design and structure of a network. Network architecture includes hardware, software, network topology, and communication protocols.

Signal transmission and encoding in Computer Networks

- Signal transmission is the process of sending data from one point to another using electromagnetic waves or electrical signals.
- Encoding is the process of converting data into a specific format that can be transmitted and recognized by the receiver.
- There are two types of signals: analog and digital.
  - Analog signals are continuous waves that vary in amplitude and frequency. They can represent any value within a range. Examples of analog signals are sound waves, light waves, and radio waves.
  - Digital signals are discrete pulses that have only two values: 0 or 1. They can represent binary data or information. Examples of digital signals are Morse code, binary code, and digital audio.
- There are two types of encoding: analog encoding and digital encoding.
  - Analog encoding is the process of modulating an analog signal with another analog signal to transmit data. There are three types of analog encoding: amplitude modulation (AM), frequency modulation

(FM), and phase modulation (PM).
    * AM is the process of varying the amplitude of a carrier signal according to the data signal. The frequency and phase of the carrier signal remain constant. AM is used for radio broadcasting, television, and telephone.
    * FM is the process of varying the frequency of a carrier signal according to the data signal. The amplitude and phase of the carrier signal remain constant. FM is used for radio broadcasting, television, and wireless communication.
    * PM is the process of varying the phase of a carrier signal according to the data signal. The amplitude and frequency of the carrier signal remain constant. PM is used for satellite communication, radar, and digital audio.
  - Digital encoding is the process of converting digital data into a digital signal that can be transmitted over a medium. There are four types of digital encoding: unipolar, polar, bipolar, and multilevel.
    * Unipolar encoding is the process of using only one voltage level to represent binary data. A high voltage represents 1 and a low voltage represents 0. Unipolar encoding is simple and cheap, but it has low bandwidth and high DC component. Unipolar encoding is used for short-distance communication and optical fiber.
    * Polar encoding is the process of using two voltage levels to represent binary data. A positive voltage represents 1 and a negative voltage represents 0. Polar encoding has higher bandwidth and lower DC component than unipolar encoding, but it has synchronization and baseline wandering issues. Polar encoding is used for long-distance communication and coaxial cable.
    * Bipolar encoding is the process of using three voltage levels to represent binary data. A positive voltage represents 1, a negative voltage represents 0, and a zero voltage represents no change. Bipolar encoding has higher bandwidth and no DC component than polar encoding, but it has more complexity and requires more voltage levels. Bipolar encoding is used for digital telephony and twisted pair cable.
    * Multilevel encoding is the process of using more than three voltage levels to represent binary data. Each voltage level represents a combination of bits. Multilevel encoding has higher bandwidth and lower noise than bipolar encoding, but it has more complexity and requires more voltage levels. Multilevel encoding is used for high-speed communication and optical fiber.

Network performance and transmission impairments in Computer Networks

- Network performance is the measure of how well a network can deliver data and services to its users. It can be evaluated by various metrics, such as throughput, delay, jitter, packet loss, availability, reliability, etc.

- Transmission impairments are the factors that degrade the quality of a signal as it travels through a medium. They can cause distortion, attenuation, noise, interference, etc. Some common transmission impairments are:
  - Attenuation: The loss of signal strength due to the resistance of the medium, the distance traveled, or the presence of obstacles. It can be measured in decibels (dB) and can be compensated by amplifiers or repeaters.
  - Distortion: The alteration of the shape or frequency of a signal due to the characteristics of the medium or the devices involved. It can cause errors or misinterpretation of the data. It can be reduced by using filters or equalizers.
  - Noise: The unwanted or random signals that are added to the original signal due to the thermal agitation of electrons, electromagnetic interference, crosstalk, etc. It can affect the signal-to-noise ratio (SNR) and the bit error rate (BER) of the transmission. It can be minimized by using shielding, modulation, error detection and correction, etc.
  - Interference: The overlapping or superposition of signals from different sources that share the same frequency band or channel. It can cause collisions or contention among the signals and reduce the network performance. It can be avoided by using frequency division multiplexing (FDM), code division multiple access (CDMA), etc.

Switching techniques and multiplexing in Computer Networks

- Switching techniques are methods of transmitting data over a network by establishing a connection between two nodes and sending data in the form of packets or messages.
- Multiplexing is a technique of combining multiple signals into one signal over a shared medium, such as a cable or a fiber optic link.
- There are three main types of switching techniques: circuit switching, message switching, and packet switching.
- Circuit switching is a technique where two nodes communicate with each other over a dedicated communication path, which is established before the data transmission and terminated after the data transmission. The path is reserved for the duration of the communication and no other nodes can use it. Circuit switching is suitable for real-time applications, such as voice and video calls, but it is inefficient for bursty data, such as web browsing and email.
- Message switching is a technique where the whole message is treated as a data unit and stored and forwarded by intermediate nodes until it reaches the destination. The message is divided into fixed-length blocks and each block is appended with a header containing the source and destination addresses. Message switching does not require a dedicated path, but it

introduces delay and overhead due to the storage and forwarding process. Message switching is suitable for applications that can tolerate delay, such as email and file transfer, but it is not suitable for real-time applications.
- Packet switching is a technique where the message is broken down into smaller chunks called packets, which are transmitted independently over the network. Each packet contains a header with the source and destination addresses, a sequence number, and a checksum. Packet switching does not require a dedicated path, but it uses statistical multiplexing to share the network resources among multiple nodes. Packet switching is suitable for both bursty and real-time applications, as it can adapt to the network congestion and provide quality of service guarantees.
- There are two methods of multiplexing multiple signals into a single channel or path or carrier: frequency division multiplexing (FDM) and time division multiplexing (TDM).
- FDM is a technique where the frequency spectrum of the channel is divided into several non-overlapping frequency bands, and each signal is modulated with a different carrier frequency and transmitted over the corresponding band. FDM is suitable for analog signals, such as voice and radio, but it is inefficient for digital signals, as it wastes bandwidth and introduces crosstalk and interference.
- TDM is a technique where the time duration of the channel is divided into several time slots, and each signal is transmitted over the channel in a round-robin fashion. TDM is suitable for digital signals, such as data and video, but it is inefficient for analog signals, as it introduces quantization error and synchronization issues.

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks

- The link layer is the lowest layer in the TCP/IP model. It is responsible for sending and receiving frames between nodes on the same physical link.
- A frame is a unit of data that contains a header, a payload, and a trailer. The header contains information such as the source and destination addresses, the type of data, and the error detection code. The payload is the actual data to be transmitted. The trailer contains a checksum or a cyclic redundancy check (CRC) to verify the integrity of the data.
- The link layer can be divided into two sublayers: the logical link control (LLC) and the medium access control (MAC). The LLC provides services such as flow control, error control, and multiplexing to the upper layers. The MAC regulates the access to the shared medium, such as a cable or a wireless channel, and prevents collisions or interference among multiple nodes.
- The link layer can use different protocols depending on the type of medium and the network topology. Some common link layer protocols are Ethernet, Wi-Fi, Bluetooth, and PPP.
- Ethernet is the most widely used link layer protocol for wired networks.

It uses a bus or a star topology, where nodes are connected to a common cable or a hub/switch. Ethernet uses a technique called carrier sense multiple access with collision detection (CSMA/CD) to avoid collisions on the shared medium. CSMA/CD works as follows:
  - A node that wants to transmit a frame first senses the medium to check if it is idle or busy. If it is idle, the node starts transmitting the frame. If it is busy, the node waits until the medium becomes idle.
  - While transmitting, the node monitors the medium for any signs of collision, such as a change in the voltage level or a garbled signal. If a collision is detected, the node stops transmitting and sends a jamming signal to notify other nodes of the collision. The node then waits for a random amount of time before trying again.
  - The random waiting time is determined by a backoff algorithm, which increases the range of possible waiting times after each collision. This reduces the probability of repeated collisions and ensures fairness among the nodes.
- Wi-Fi is the most widely used link layer protocol for wireless networks. It uses a star topology, where nodes are connected to a central device called an access point (AP). Wi-Fi uses a technique called carrier sense multiple access with collision avoidance (CSMA/CA) to avoid collisions on the shared medium. CSMA/CA works as follows:
  - A node that wants to transmit a frame first senses the medium to check if it is idle or busy. If it is idle, the node sends a short control frame called a request to send (RTS) to the AP, indicating the duration of the transmission. The AP replies with a short control frame called a clear to send (CTS), granting the permission to the node and reserving the medium for the specified duration. The node then transmits the frame and waits for an acknowledgment (ACK) from the AP. If the ACK is received, the transmission is successful. If the ACK is not received, the node assumes a collision or a loss and tries again after a random waiting time.
  - If the medium is busy, the node waits until the medium becomes idle and then starts a countdown timer called a backoff timer. The backoff timer is initialized with a random value from a contention window, which is a range of possible waiting times. The node decrements the backoff timer as long as the medium is idle, and pauses the timer if the medium is busy. When the backoff timer reaches zero, the node sends the RTS and proceeds as above.
  - The contention window is determined by a backoff algorithm, which increases the range of possible waiting times after each unsuccessful transmission. This reduces the probability of repeated collisions and ensures fairness among the nodes.
- Bluetooth is a link layer protocol for short-range wireless communication among devices such as smartphones, laptops, headphones, and speakers. It uses a technique called frequency hopping spread spectrum (FHSS) to

avoid interference on the shared medium. FHSS works as follows:

- The wireless spectrum is divided into 79 channels, each with a bandwidth of 1 MHz. The devices use a pseudo-random sequence to hop from one channel to another at a rate of 1600 hops per second. This makes it unlikely that two devices will use the same channel at the same time, and reduces the impact of noise or interference on any single channel.
- The devices form a network called a piconet, which consists of one master device and up to seven active slave devices. The master device coordinates the channel hopping and the communication among

Link layer in Computer Networks

- The link layer is the lowest layer in the Internet protocol suite, the networking architecture of the Internet.
- The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to.
- The link layer is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer may also provide the means to detect and possibly correct errors that can occur in the physical layer.
- The link layer is concerned with local delivery of frames between nodes on the same level of the network.
- The link layer can be divided into two sublayers: data link control and multiple access resolution/protocol.
- Data link control is the sublayer that handles framing, addressing, error control, flow control, and media access control.
- Multiple access resolution/protocol is the sublayer that handles the coordination of multiple nodes sharing the same medium, such as a wireless channel or a bus network.
- Some examples of multiple access protocols are ALOHA, CSMA, CSMA/CA, and CSMA/CD.

Framing in link layer in Computer Networks

- Framing is a function of the data link layer that provides a way for a sender to transmit a set of bits that are meaningful to the receiver .
- Framing uses frames to send or receive data. A frame is a protocol data unit at the data link layer that consists of a link layer header followed by a packet.
- The link layer header contains information such as source and destination addresses, error-checking codes, and control information.
- Framing is necessary because the physical layer only accepts and transfers a stream of bits without any regard to meaning or structure.
- Framing also helps to synchronize the sender and receiver by marking the start and end of each frame.

- There are different types of framing methods, such as character-oriented, bit-oriented, and clock-based framing .
- Character-oriented framing uses special characters to indicate the start and end of each frame, such as STX (start of text) and ETX (end of text). This method is simple but may encounter problems if the data contains the same special characters.
- Bit-oriented framing uses a special bit pattern, such as 01111110, to indicate the start and end of each frame. This method is more efficient but requires bit stuffing to avoid confusion if the data contains the same bit pattern.
- Clock-based framing uses a clock signal to synchronize the sender and receiver and to determine the frame boundaries. This method is reliable but requires a dedicated channel for the clock signal.

Error Detection and Correction in link layer in Computer Networks

- Error detection and correction are techniques that allow reliable data transmission over noisy communication channels, such as wired or wireless networks.
- Error detection is the process of identifying corrupted bits or frames in the transmitted data, while error correction is the process of recovering the original data from the corrupted bits or frames.
- Error detection and correction can be performed at different layers of the network stack, but the link layer is the most common layer where these techniques are applied.
- The link layer is responsible for transferring data frames between adjacent nodes in a network, such as routers, switches, or hosts.
- The link layer can use different error detection and correction schemes, depending on the characteristics of the underlying physical medium, the data rate, the error rate, and the performance requirements.
- Some of the common error detection and correction schemes used in the link layer are:
  - Parity check: A simple technique that adds a single bit to each data unit, such that the number of 1s in the data unit and the parity bit is even (even parity) or odd (odd parity). The receiver can detect a single bit error by checking the parity of the received data unit, but cannot correct it.
  - Checksum: A technique that computes a fixed-length value, called a checksum, from the data unit, and appends it to the data unit. The receiver can compute the checksum of the received data unit and compare it with the appended checksum to detect errors. The checksum can detect multiple bit errors, but cannot correct them.

- Cyclic redundancy check (CRC): A technique that generates a fixed-length value, called a CRC, from the data unit, using a predefined polynomial function, and appends it to the data unit. The receiver can use the same polynomial function to compute the CRC of the received data unit and compare it with the appended CRC to detect errors. The CRC can detect burst errors, which are errors that affect a sequence of consecutive bits, but cannot correct them.
- Hamming code: A technique that adds extra bits, called parity bits, to the data unit, such that any single bit error can be detected and corrected by the receiver. The number and position of the parity bits are determined by the Hamming distance, which is the minimum number of bit changes required to convert one valid code word into another. The receiver can use the parity bits to locate and correct the erroneous bit in the received data unit.
- Reed-Solomon code: A technique that encodes the data unit into a longer code word, such that multiple bit errors can be detected and corrected by the receiver. The code word consists of symbols, which are groups of bits, and the encoding and decoding are based on polynomial arithmetic over finite fields. The receiver can use the redundancy in the code word to correct the corrupted symbols in the received data unit.

Flow control in link layer in Computer Networks

- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It regulates the amount of data that a sender can send before receiving an acknowledgment from the receiver .
- It prevents the sender from overwhelming the receiver with too many frames or data packets.
- There are two main methods of flow control in the link layer: stop-and-wait and sliding window.
- Stop-and-wait is a simple method where the sender sends one frame at a time and waits for an acknowledgment from the receiver before sending the next frame.
- Sliding window is a more efficient method where the sender can send multiple frames without waiting for acknowledgments, as long as the number of frames does not exceed the window size agreed by both stations.
- The window size is the number of frames that can be sent or received at a time.
- The sender maintains a send window that indicates the range of frames that can be sent, and the receiver maintains a receive window that indicates the range of frames that can be accepted.
- The sender and the receiver update their windows based on the acknowledgments and feedback they receive from each other.
- Flow control can also be implemented at the Ethernet level using pause

frames, which are special frames that can be sent by the receiver to the sender to temporarily stop the transmission of data.

Elementary Data Link Protocols in link layer in Computer Networks

- The data link layer is responsible for framing, error control and flow control in the communication between two adjacent nodes in a network.
- Framing is the process of dividing the bit stream from the physical layer into data frames whose size ranges from a few hundred to a few thousand bytes.
- Error control is the process of detecting and correcting errors that may occur during transmission or reception of data frames.
- Flow control is the process of regulating the rate of data transmission between the sender and the receiver to avoid congestion or buffer overflow.
- There are different types of protocols in the data link layer that perform these functions in different ways. Some of the elementary data link protocols are:
  - Protocol 1: Unrestricted simplex protocol
    * This protocol assumes that there is a one-way communication channel between the sender and the receiver, and that there are no errors or losses in the channel.
    * The sender simply sends data frames continuously without waiting for any acknowledgment or feedback from the receiver.
    * The receiver accepts and processes the incoming data frames as fast as possible.
    * This protocol is simple and efficient, but it does not provide any error or flow control mechanisms.
    * This protocol is suitable for applications that do not require reliable or bidirectional communication, such as broadcasting or streaming.
  - Protocol 2: Simplex stop-and-wait protocol
    * This protocol assumes that there is a one-way communication channel between the sender and the receiver, but that there may be errors or losses in the channel.
    * The sender sends one data frame at a time and waits for an acknowledgment from the receiver before sending the next frame.
    * The receiver sends an acknowledgment for each received frame after checking for errors. If the frame is corrupted or lost, the receiver does not send any acknowledgment and discards the frame.
    * The sender uses a timer to detect the loss of acknowledgment and retransmits the frame if the timer expires.
    * This protocol provides error control, but it does not provide flow

control. It also suffers from low efficiency and long delay due to the waiting time between frames.

- Protocol 3: Simplex protocol for noisy channels
  * This protocol assumes that there is a one-way communication channel between the sender and the receiver, and that there may be errors or losses in the channel. It also assumes that the channel may introduce duplicate frames due to noise or retransmission.
  * The sender sends one data frame at a time and waits for an acknowledgment from the receiver before sending the next frame. The sender also adds a sequence number to each frame to distinguish between original and duplicate frames.
  * The receiver sends an acknowledgment for each received frame after checking for errors and sequence number. If the frame is corrupted, lost or duplicated, the receiver does not send any acknowledgment and discards the frame.
  * The sender uses a timer to detect the loss of acknowledgment and retransmits the frame if the timer expires.
  * This protocol provides error control and handles duplicate frames, but it does not provide flow control. It also suffers from low efficiency and long delay due to the waiting time between frames.

Sliding Window protocols in link layer in Computer Networks

- The sliding window protocol is a data link layer protocol that is useful in the sequential and reliable delivery of the data frames .
- Using the sliding window protocol, the sender can send multiple frames at a time before receiving an acknowledgment from the receiver .
- The sliding window is also used in Transmission Control Protocol (TCP), which operates at the transport layer .
- The sliding window protocol manages the flow of data between two network nodes to ensure that the receiver can handle the incoming data .
- The sliding window protocol uses two types of windows: sender window and receiver window .
- The sender window is the set of frames that the sender can send without waiting for an acknowledgment .
- The receiver window is the set of frames that the receiver can accept without sending an acknowledgment .
- The size of the windows can vary depending on the protocol and the network conditions .
- The sliding window protocol can be classified into three types: stop-and-wait, go-back-N, and selective repeat .
- In stop-and-wait, the sender sends one frame at a time and waits for an acknowledgment before sending the next frame .
- In go-back-N, the sender can send up to N frames at a time and waits for

an acknowledgment for the first frame before sliding the window .

- In selective repeat, the sender can send up to N frames at a time and waits for an acknowledgment for each frame before sliding the window .
- The sliding window protocol improves the efficiency and reliability of data transmission by reducing the waiting time and handling the errors and losses .

Medium Access Control and Local Area Networks

- Medium Access Control (MAC) is a sublayer of the data link layer that coordinates the access of multiple devices to a shared communication medium, such as a cable, a wireless channel, or a bus.
- MAC protocols are designed to avoid or resolve collisions, which occur when two or more devices transmit data at the same time on the same medium, resulting in garbled signals and data loss.
- MAC protocols can be classified into two main categories: contention-based and reservation-based.
  - Contention-based protocols allow devices to compete for the medium access, using random or deterministic methods to resolve collisions. Examples of contention-based protocols are ALOHA, Carrier Sense Multiple Access (CSMA), and CSMA with Collision Detection (CSMA/CD).
  - Reservation-based protocols allocate the medium access to devices in advance, using polling, token passing, or time division multiplexing (TDM) methods to avoid collisions. Examples of reservation-based protocols are Token Ring, Token Bus, and Time Division Multiple Access (TDMA).
- Local Area Networks (LANs) are networks that connect devices within a limited geographic area, such as a building, a campus, or a home.
- LANs typically use MAC protocols to coordinate the access of devices to a shared medium, such as Ethernet, Wi-Fi, or Bluetooth.
- LANs can be characterized by their topology, which is the physical or logical arrangement of devices and links in the network. Common LAN topologies are bus, ring, star, tree, and mesh.

Channel allocation in medium access control

- Channel allocation is the process of assigning channels (frequency bands, time slots, code sequences, etc.) to different users or applications that share a common communication medium.
- Medium access control (MAC) is the sublayer of the data link layer that coordinates the access to the shared medium among multiple MAC entities (nodes, devices, etc.).
- Channel allocation and medium access control are closely related, as the MAC protocol determines how the channels are allocated and accessed by the MAC entities.

- There are different types of channel allocation schemes, such as:
  - Fixed channel allocation: each channel is assigned to a specific user or application and remains fixed for the duration of the communication. This scheme is simple and efficient, but may result in underutilization or wastage of channels if the traffic is not uniform or predictable.
  - Dynamic channel allocation: channels are assigned to users or applications on demand, based on the current traffic conditions and channel availability. This scheme is more flexible and adaptive, but may incur higher overhead and complexity in channel management and coordination.
  - Hybrid channel allocation: a combination of fixed and dynamic channel allocation, where some channels are reserved for specific users or applications, while others are available for dynamic allocation. This scheme can balance the trade-off between efficiency and flexibility, but may require more sophisticated MAC protocols and algorithms.
- There are different types of MAC protocols, such as:
  - Contention-based MAC: MAC entities compete for the access to the channel, using random or deterministic methods to resolve the contention. This type of MAC is suitable for dynamic and bursty traffic, but may result in collisions, delays, and unfairness.
  - Reservation-based MAC: MAC entities reserve the channel in advance, using a centralized or distributed mechanism to coordinate the reservations. This type of MAC is suitable for predictable and periodic traffic, but may result in inefficiency and overhead if the traffic is variable or unpredictable.
  - Hybrid MAC: a combination of contention-based and reservation-based MAC, where some channels are accessed by contention, while others are accessed by reservation. This type of MAC can adapt to different traffic patterns and requirements, but may require more complex MAC protocols and algorithms.

Multiple access protocols in medium access control

Multiple access protocols are a set of protocols operating in the Medium Access Control sublayer (MAC sublayer) of the Open Systems Interconnection (OSI) model. These protocols allow a number of nodes or users to access a shared network channel. The main objectives of multiple access protocols are to avoid collisions, minimize delay, maximize throughput, and ensure fairness.

There are three main categories of multiple access protocols:

- **Random access protocols**: In these protocols, all stations have equal priority and can send data depending on the state of the medium (idle or busy). There is no fixed time for sending data and collisions may occur. Examples of random access protocols are ALOHA, Carrier Sense Multiple Access (CSMA), CSMA with Collision Avoidance (CSMA/CA), and CSMA with Collision Detection (CSMA/CD).

- **Controlled access protocols**: In these protocols, a station needs to obtain permission from a central authority or follow a predefined order before sending data. There is a fixed time for sending data and collisions are avoided. Examples of controlled access protocols are Reservation, Polling, and Token Passing.
- **Channelization protocols**: In these protocols, the available bandwidth of the channel is divided into smaller units and assigned to different stations. There is no fixed time for sending data and collisions are avoided. Examples of channelization protocols are Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA), and Orthogonal Frequency Division Multiple Access (OFDMA).

LAN standards in local area network

- A local area network (LAN) is a data communication network connecting various terminals or computers within a building or limited geographical area. The connection among the devices could be wired or wireless.
- LAN standards are the specifications and protocols that define the operation and performance of a LAN. They cover aspects such as the physical layer, the data link layer, the network layer, and the application layer of the network.
- The most widely used family of LAN standards is the IEEE 802, which consists of twelve members, numbered 802.1 through 802.12, with a focus group of the LMSC (LAN/MAN Standards Committee) devoted to each .
- Some of the most common IEEE 802 standards are:
  - 802.3: Ethernet, the dominant LAN technology that uses a bus or star topology and supports data rates of 10 Mbps to 100 Gbps.
  - 802.11: Wireless LAN (WLAN), which uses radio waves to provide wireless access to a LAN. It supports data rates of up to 600 Mbps and has several variants such as 802.11a, 802.11b, 802.11g, 802.11n, and 802.11ac.
  - 802.15: Wireless Personal Area Network (WPAN), which enables short-range wireless communication among devices such as smartphones, tablets, laptops, and printers. It includes standards such as Bluetooth, ZigBee, and NFC.
  - 802.16: Wireless Metropolitan Area Network (WMAN), which provides wireless broadband access to a metropolitan area. It is also known as WiMAX and can support data rates of up to 1 Gbps.
- Other LAN standards that are not part of the IEEE 802 family include:
  - Token Ring, which uses a ring topology and a token-passing mechanism to regulate access to the network. It supports data rates of 4 Mbps or 16 Mbps.
  - Fiber Distributed Data Interface (FDDI), which uses optical fiber as the transmission medium and a dual ring topology to provide high-speed and reliable communication. It supports data rates of 100

Mbps.
  - Asynchronous Transfer Mode (ATM), which uses fixed-length cells to transmit data over various types of media. It can support data rates of up to 622 Mbps and can integrate voice, video, and data on the same network.

Link layer switches & bridges in local area network

- Link layer switches and bridges are network devices that operate at the data link layer (layer 2) of the OSI model.
- They connect multiple LANs (local area networks) together to form a larger LAN or a single broadcast domain.
- They use MAC addresses to forward Ethernet frames from one device to another device in an Ethernet standard based LAN.
- They can also interconnect data link layer domains that have different technologies, such as Ethernet to FDDI, by converting the frame formats and adjusting the maximum frame size.
- They store and forward the frames, which means they receive the entire frame, check for errors, and then transmit it to the destination or drop it if there is no destination.
- They can learn the MAC addresses of the devices connected to their ports by examining the source address of the incoming frames, and build a MAC address table to store the mappings of MAC addresses and ports.
- They can filter the frames by only forwarding them to the ports that belong to the destination MAC address, or broadcast them to all ports if the destination MAC address is unknown or multicast.
- They can improve the network performance by reducing the collision domain and increasing the bandwidth for each device.
- They can also perform some advanced functions, such as VLANs, spanning tree protocol, link aggregation, and quality of service.

Learning bridge algorithms in local area network

- A bridge is a device that connects two or more local area networks (LANs) and forwards data frames between them based on their MAC addresses.
- A bridge algorithm is a set of rules that determines how a bridge learns the MAC addresses of the devices connected to the LANs and how it decides which frames to forward or filter.
- There are different types of bridge algorithms, such as transparent bridging, source routing bridging, and spanning tree protocol.
- Transparent bridging is the most common type of bridge algorithm. It uses two main functions: learning and filtering/forwarding.
  - Learning: The bridge maintains a table that maps each MAC address to the port where it was last seen. Whenever the bridge receives a frame, it updates the table with the source MAC address and the port number.

  - Filtering/forwarding: The bridge checks the destination MAC address of each frame against the table. If the destination MAC address is found in the table and the port number matches the incoming port, the bridge filters (discards) the frame. If the destination MAC address is not found in the table or the port number does not match the incoming port, the bridge forwards the frame to all other ports (flooding).
- Source routing bridging is another type of bridge algorithm that is used in token ring networks. It relies on the source device to specify the path of the frame through the network using a routing information field (RIF) in the frame header.
  - The RIF contains a list of bridge numbers and ring numbers that the frame must traverse to reach the destination.
  - The bridge checks the RIF and forwards the frame to the appropriate port based on the next bridge number and ring number in the list.
  - The bridge also updates the RIF with its own bridge number and ring number as the frame passes through it.
- Spanning tree protocol (STP) is a bridge algorithm that prevents loops in a network with multiple bridges. It creates a logical tree structure of the network by selecting a root bridge and designating ports as either root ports, designated ports, or blocked ports.
  - Root bridge: The bridge with the lowest bridge ID (a combination of priority and MAC address) in the network. It is the root of the spanning tree.
  - Root port: The port on each non-root bridge that has the lowest cost path to the root bridge. It is used to forward frames to and from the root bridge.
  - Designated port: The port on each LAN segment that has the lowest cost path to the root bridge. It is used to forward frames to and from the LAN segment.
  - Blocked port: The port on each bridge that is neither a root port nor a designated port. It is not used to forward frames and is in a listening state.
  - The bridge algorithm uses a protocol called bridge protocol data units (BPDUs) to exchange information about the bridge IDs, port costs, and port states among the bridges and to elect the root bridge and the designated ports.

Spanning Tree Algorithms in Local Area Network

- Spanning tree algorithms are used to prevent loops in a network topology that uses bridges or switches to connect multiple segments of a local area network (LAN).
- Loops can cause problems such as broadcast storms, multiple frame transmission, and MAC address table instability.
- Spanning tree algorithms work by creating a logical tree structure of the

network, where only one path exists between any two nodes. This is done by blocking some of the links that create loops, while keeping others as backup links in case of link failure.

- The most common spanning tree algorithm is the Spanning Tree Protocol (STP), which is standardized by IEEE 802.1D. STP operates as follows :
  - STP elects one switch as the root bridge, which is the central point of the spanning tree. The root bridge is chosen based on the lowest bridge ID, which is a combination of a priority value and a MAC address.
  - STP assigns a cost to each link based on the bandwidth of the link. The lower the bandwidth, the higher the cost.
  - STP calculates the shortest path from each switch to the root bridge, based on the sum of the link costs. This path is called the root port for each switch.
  - STP determines the best link to forward traffic between switches on the same segment, based on the lowest port ID, which is a combination of a priority value and a port number. This link is called the designated port for each segment.
  - STP blocks all other links that are not root ports or designated ports. These links are called non-designated ports or alternate ports.
  - STP periodically sends special frames called Bridge Protocol Data Units (BPDUs) to exchange information about the network topology and detect changes. BPDUs are sent from the root bridge to all other switches, and from each designated port to its segment.
  - STP adapts to changes in the network topology by recalculating the spanning tree and unblocking or blocking ports as needed. This process is called convergence and may take several seconds to complete.

Unit 3 - Network Layer in Computer Networks

The network layer is the third layer of the OSI model and the TCP/IP model. It is responsible for the following functions:

- Routing: The process of finding a path from a source to a destination in the network.
- Addressing: The process of assigning a unique identifier to each device in the network.
- Packetization: The process of dividing a large message into smaller units called packets.
- Fragmentation and reassembly: The process of breaking a packet into smaller fragments and reassembling them at the destination.
- Error control: The process of detecting and correcting errors in the transmission of packets.
- Congestion control: The process of avoiding or reducing the overload of the network resources.
- Quality of service: The process of providing different levels of service to

different types of traffic.

Some of the common protocols and devices used in the network layer are:

- IP: Internet Protocol, the main protocol for addressing and routing packets in the Internet.
- ICMP: Internet Control Message Protocol, a protocol for sending error and control messages in the network layer.
- ARP: Address Resolution Protocol, a protocol for mapping a network layer address to a data link layer address.
- RARP: Reverse Address Resolution Protocol, a protocol for mapping a data link layer address to a network layer address.
- DHCP: Dynamic Host Configuration Protocol, a protocol for assigning network layer addresses and other configuration parameters to hosts dynamically.
- NAT: Network Address Translation, a technique for translating a private network layer address to a public one and vice versa.
- RIP: Routing Information Protocol, a distance vector routing protocol for exchanging routing information among routers.
- OSPF: Open Shortest Path First, a link state routing protocol for exchanging routing information among routers.
- BGP: Border Gateway Protocol, a path vector routing protocol for exchanging routing information among autonomous systems.
- Router: A device that connects two or more networks and forwards packets based on their network layer addresses.

Point-to-point networks in network layer

- A point-to-point network is a network topology that connects two nodes directly using a single link.
- A point-to-point network can be either physical or logical. A physical point-to-point network uses a dedicated cable or wireless channel to connect the nodes. A logical point-to-point network uses a shared medium, such as a bus or a ring, but assigns a unique address to each node and uses a protocol to ensure that only one node can transmit at a time.
- A point-to-point network can be either simplex, half-duplex, or full-duplex. A simplex point-to-point network allows only one node to transmit and the other to receive. A half-duplex point-to-point network allows both nodes to transmit and receive, but not at the same time. A full-duplex point-to-point network allows both nodes to transmit and receive simultaneously.
- A point-to-point network can be either synchronous or asynchronous. A synchronous point-to-point network uses a common clock signal to synchronize the transmission and reception of data. An asynchronous point-to-point network does not use a clock signal, but relies on start and stop bits to mark the beginning and end of each data unit.
- A point-to-point network can be either circuit-switched or packet-switched. A circuit-switched point-to-point network establishes a dedicated connec-

tion between the nodes for the duration of the communication. A packet-switched point-to-point network divides the data into smaller units called packets and sends them over the link independently, possibly using different routes.

- A point-to-point network can be either unicast, multicast, or broadcast. A unicast point-to-point network sends data from one node to another node. A multicast point-to-point network sends data from one node to a subset of nodes. A broadcast point-to-point network sends data from one node to all nodes.
- A point-to-point network can be either connection-oriented or connectionless. A connection-oriented point-to-point network requires the nodes to establish a session before exchanging data. A connectionless point-to-point network does not require a session, but sends data as datagrams without any guarantee of delivery, order, or error correction.
- A point-to-point network can be either reliable or unreliable. A reliable point-to-point network ensures that the data is delivered correctly and in order, using mechanisms such as acknowledgment, retransmission, and checksum. An unreliable point-to-point network does not provide any guarantee of data delivery, order, or error correction, and may lose, duplicate, or corrupt data.
- A point-to-point network can be either secure or insecure. A secure point-to-point network protects the data from unauthorized access, modification, or disclosure, using techniques such as encryption, authentication, and digital signature. An insecure point-to-point network does not provide any security for the data, and may expose it to eavesdropping, tampering, or impersonation.

Logical addressing in network layer

- Logical addressing is a way of identifying devices on a network using addresses that are assigned by a network layer protocol, such as IP or IPX .
- Logical addresses are also called network addresses or layer 3 addresses.
- Logical addresses are independent of the physical addresses, such as MAC addresses, that are used by the data link layer or layer 2 .
- Logical addresses allow devices to communicate across different physical networks, such as LANs or WANs, by using routers that can translate logical addresses to physical addresses .
- Logical addresses are usually hierarchical, meaning they have two parts: a network part and a host part. The network part identifies the network to which the device belongs, and the host part identifies the device within that network.
- Logical addresses are usually represented in a human-readable format, such as dotted decimal notation for IPv4 addresses or hexadecimal notation for IPv6 addresses .
- Logical addresses are placed in the header of the packets that are trans-

mitted by the network layer. The sender and receiver's logical addresses are used to route the packets to their destination.

- Logical addresses are subject to change if the device moves to a different network or if the network layer protocol is changed .

Basic internetworking in network layer

- The network layer is responsible for routing packets across different networks or subnets.
- The network layer uses logical addresses, such as IP addresses, to identify the source and destination of each packet.
- The network layer relies on the data link layer to deliver packets within the same network or subnet, and on the transport layer to ensure reliable and ordered delivery of packets across different networks or subnets.
- The network layer can use different protocols to perform routing, such as RIP, OSPF, EIGRP, BGP, etc.
- The network layer can also provide other functions, such as fragmentation, reassembly, congestion control, quality of service, etc.
- The network layer is the third layer of the OSI model and the second layer of the TCP/IP model.

IP

- IP stands for Internet Protocol, which is a set of rules that governs how data packets are transmitted across a network.
- IP is one of the main protocols in the TCP/IP suite, which is a collection of protocols and standards that enable communication between different devices and applications on the Internet.
- IP has two main functions: addressing and routing.
  - Addressing: IP assigns a unique numerical address to each device on the network, called an IP address. An IP address consists of four numbers separated by dots, such as 192.168.1.1. Each number can range from 0 to 255. IP addresses are used to identify the source and destination of data packets.
  - Routing: IP determines the best path for data packets to travel from the source to the destination, based on factors such as distance, congestion, and availability. IP uses routers, which are devices that forward data packets between different networks, to perform routing. IP also handles the fragmentation and reassembly of data packets, which are divided into smaller units for transmission and recombined at the destination.
- IP is a connectionless and unreliable protocol, which means that it does not establish a direct connection between the source and destination, and it does not guarantee the delivery, order, or integrity of data packets. IP relies on other protocols, such as TCP, to provide these features.
- IP has two versions: IPv4 and IPv6.

- IPv4: IPv4 is the most widely used version of IP, which uses 32-bit addresses, allowing for about 4.3 billion possible addresses. However, due to the rapid growth of the Internet, IPv4 addresses are running out, and new devices and applications require more addresses than IPv4 can provide.
- IPv6: IPv6 is the newer version of IP, which uses 128-bit addresses, allowing for about 3.4 x 10^38 possible addresses. IPv6 also offers other advantages over IPv4, such as improved security, efficiency, and scalability. However, IPv6 is not fully compatible with IPv4, and the transition from IPv4 to IPv6 is still ongoing.

CIDR

- CIDR stands for Classless Inter-Domain Routing, a method for allocating IP addresses and for IP routing .
- CIDR replaces the previous classful network addressing architecture on the Internet, which was based on fixed classes of IP addresses (A, B, and C) .
- CIDR allows blocks of IP addresses to be grouped into single routing table entries, which reduces the size and complexity of routing tables and improves the efficiency of address distribution .
- CIDR is based on a bitwise, prefix-based representation of IP addresses and their routing properties. For example, 192.168.0.0/24 is a CIDR notation that represents the IP address range from 192.168.0.0 to 192.168.0.255, where the /24 indicates that the first 24 bits of the address are fixed and the remaining 8 bits can vary .
- CIDR also enables the use of variable-length subnet masks (VLSM), which allow subnets to be divided into smaller subnets of different sizes, depending on the network requirements.
- CIDR is an important technique for managing the limited address space of IPv4, the current version of the Internet Protocol. However, CIDR is also compatible with IPv6, the next generation of the Internet Protocol, which has a much larger address space and supports more hierarchical levels of routing.

ARP

- ARP stands for Address Resolution Protocol, which is a communication protocol used for discovering the link layer address, such as a MAC address, associated with a given internet layer address, typically an IPv4 address .
- ARP is a critical function in the Internet protocol suite, as it enables devices to communicate within a local area network (LAN) by mapping IP addresses to MAC addresses .
- ARP works by sending broadcast messages to all devices in the LAN, asking for the MAC address of the device that owns a specific IP address. The device that matches the IP address replies with its MAC address, and

the sender stores this information in a cache for future use.

- ARP has two types of messages: ARP request and ARP reply. An ARP request is a message that asks for the MAC address of a device with a given IP address. An ARP reply is a message that provides the MAC address of a device with a given IP address.
- ARP has some limitations and vulnerabilities, such as:
  - ARP does not verify the identity of the sender or the receiver, which makes it susceptible to spoofing and poisoning attacks.
  - ARP does not support IPv6 addresses, which require a different protocol called Neighbor Discovery Protocol (NDP).
  - ARP can cause network congestion and performance degradation if there are too many ARP requests and replies in the LAN.

RARP

- RARP stands for Reverse Address Resolution Protocol.
- It is a protocol based on computer networking that is used by a client computer to request its IP address from a gateway server's Address Resolution Protocol (ARP) table or cache.
- It works by sending the device's physical address (MAC address) to a specialized RARP server that is on the same local area network (LAN) and is actively listening for RARP requests.
- The RARP server then looks up the MAC address in its ARP table and sends back the corresponding IP address to the client computer.
- RARP is on the Network Access Layer (the lowest layer of the TCP/IP protocol stack) and is thus a protocol used to send data between two points in a network.
- RARP was published in 1984 and was included in the TCP/IP protocol stack, but it is now obsolete and replaced by other protocols such as BOOTP and DHCP.

DHCP

- DHCP stands for Dynamic Host Configuration Protocol. It is a network management protocol that automatically assigns IP addresses and other communication parameters to devices connected to a network using a client-server architecture .
- DHCP enables a host to obtain an IP address dynamically without manual configuration. This simplifies the network administration and reduces the risk of IP address conflicts .
- DHCP operates on four basic steps: discover, offer, request, and acknowledge. A DHCP client sends a broadcast message to discover a DHCP server on the network. A DHCP server responds with an offer message that contains an available IP address and other configuration information. The client then sends a request message to accept the offer. The server finally sends an acknowledge message to confirm the lease of the IP address

to the client .

- DHCP can also provide other information to the client, such as the subnet mask, default gateway, domain name, DNS servers, and time servers. These are called DHCP options and are specified in RFC 2132 .
- DHCP is an IETF standard based on an earlier protocol called Bootstrap Protocol (BOOTP). DHCP is defined in RFCs 2131 and 2132. DHCP can coexist with BOOTP clients and servers on the same network.
- DHCP can support both static and dynamic allocation of IP addresses. Static allocation means that a client always receives the same IP address from the DHCP server. Dynamic allocation means that a client receives an IP address for a limited period of time, called a lease. The client must renew the lease before it expires or request a new IP address .
- DHCP is widely used in various types of networks, such as LANs, WANs, wireless networks, and VPNs. DHCP can also support IPv6 addresses, which are defined in RFC 3315.

ICMP

- ICMP stands for Internet Control Message Protocol.
- ICMP is a network layer protocol that is used by network devices to communicate problems with data transmission.
- ICMP messages are typically used for diagnostic or control purposes or generated in response to errors in IP operations.
- ICMP messages have a header and a payload. The header contains the type, code, and checksum of the message. The payload contains additional information, such as the IP header and the first 8 bytes of the original datagram that caused the error.
- Some common types of ICMP messages are:
  - Echo request and echo reply: used to test the reachability and round-trip time of a destination.
  - Destination unreachable: used to inform the source that the destination or the route to the destination is unreachable for some reason.
  - Time exceeded: used to inform the source that the datagram has expired in transit due to a hop limit or a reassembly timeout.
  - Parameter problem: used to inform the source that the datagram has an invalid or missing field in the IP header.
  - Source quench: used to inform the source that the destination or an intermediate router is experiencing congestion and requests the source to reduce its sending rate.
  - Redirect: used to inform the source that there is a better route to the destination and suggests the source to use a different gateway.
  - Timestamp request and timestamp reply: used to measure the time difference between the source and the destination.
  - Address mask request and address mask reply: used to obtain the subnet mask of a destination.

Routing in network layer

- Routing is the process of finding and selecting the best path for data packets to reach their destination in a network.
- Routing is performed by a special device known as a router, which works at the network layer in the OSI model and internet layer in TCP/IP model.
- A router forwards the packets based on the information available in the packet header and forwarding table, which contains the mapping of network addresses and outgoing interfaces.
- Routing can be based on static tables that are rarely changed, or dynamic algorithms that update the tables depending on network conditions.
- Routing can be classified into two types: intra-domain routing and inter-domain routing.
- Intra-domain routing is the routing within a single network or autonomous system (AS), which is a group of networks under the same administrative control. Intra-domain routing protocols are also called interior gateway protocols (IGPs).
- Inter-domain routing is the routing between different networks or autonomous systems. Inter-domain routing protocols are also called exterior gateway protocols (EGPs).
- Some examples of intra-domain routing protocols are RIP, OSPF, and EIGRP. Some examples of inter-domain routing protocols are BGP, EGP, and IDRP.

Forwarding and Delivery in Network Layer

- The network layer supervises the handling of packets by the underlying physical networks. We call this handling as the delivery of packets to the destination.
- The delivery of a packet is called **direct** if the deliverer (host or router) and the destination are on the same network; the delivery of a packet is called **indirect** if the deliverer (host or router) and the destination are on different networks.
- Forwarding means placing the packet in its route to the destination and it requires a routing table. A routing table is a data structure that stores the information about the routes to the destination networks.
- Forwarding refers to the router-local action of transferring a packet from an input link interface to the appropriate output link interface. Routing refers to the network-wide process that determines the end-to-end paths that packets take from source to destination.
- Address aggregation is a technique that reduces the size of the routing table by grouping several networks into a single entry. For example, if a router has four networks with the same prefix, it can aggregate them into one entry with a shorter prefix length.
- There are some tools and utilities that can help with packet delivery and routing, such as ping, traceroute, ipconfig, netstat, and route. These

tools can test the connectivity, display the route, show the configuration, monitor the traffic, and modify the routing table.

Static and dynamic routing in computer networks

- Static routing and dynamic routing are two methods used to determine how to send a packet toward its destination.
- Static routes are configured in advance of any network communication by a network administrator .
- Dynamic routes are learned by routers through exchanging information with other routers .
- Static routing uses a single preconfigured route to send traffic to its destination, while dynamic routing uses complex routing algorithms to select the best route based on network conditions .
- Static routing provides more security and control over the network traffic, but requires manual reconfiguration if the network topology changes .
- Dynamic routing provides more flexibility and scalability, but consumes more bandwidth and CPU resources to maintain routing tables .
- Static routing is often used for small networks or as a backup for dynamic routing, while dynamic routing is often used for large networks or to adapt to changing network situations .
- Types of static routes include standard static route, default static route, summary static route, and floating static route.
- Types of dynamic routing protocols include distance vector protocols, link state protocols, and hybrid protocols.

Routing algorithms and protocols in computer networks

- Routing is the process of finding and selecting the best path for data transmission from source to destination in a computer network.
- Routing algorithms are the software programs that implement the logic of routing, i.e., they decide the optimal path for each packet based on some criteria or metric.
- Routing protocols are the set of rules and procedures that routers use to exchange routing information and update their routing tables.
- Routing algorithms and protocols can be classified into different types based on various factors, such as:
  - Static vs dynamic: Static routing algorithms are fixed and do not change with the network topology or traffic conditions. Dynamic routing algorithms are adaptive and update their routing decisions based on the current network state.
  - Centralized vs distributed: Centralized routing algorithms are executed by a single node that has the complete knowledge of the network topology and traffic. Distributed routing algorithms are executed by each node independently based on the local information and messages from neighboring nodes.

  - Single-path vs multipath: Single-path routing algorithms select only one path for each source-destination pair. Multipath routing algorithms can select multiple paths for load balancing, fault tolerance, or quality of service.
  - Flat vs hierarchical: Flat routing algorithms treat all nodes as equal and use the same addressing scheme for the whole network. Hierarchical routing algorithms divide the network into regions or domains and use different levels of addressing and routing for each level.
  - Intra-domain vs inter-domain: Intra-domain routing algorithms are used within a single network domain or autonomous system, which is a group of nodes under the same administrative control. Inter-domain routing algorithms are used between different network domains or autonomous systems, which may have different policies and preferences.
- Some examples of routing protocols are:
  - Distance vector protocols: These protocols use the distance (or hop count) to each destination as the metric and exchange their routing tables with their direct neighbors periodically or when there is a change. They use the Bellman-Ford algorithm to calculate the shortest paths. Examples are RIP, IGRP, EIGRP, etc.
  - Link state protocols: These protocols use the cost of each link as the metric and exchange their link state information with all other nodes in the network periodically or when there is a change. They use the Dijkstra's algorithm to calculate the shortest paths. Examples are OSPF, IS-IS, etc.
  - Path vector protocols: These protocols use the path (or sequence of autonomous systems) to each destination as the metric and exchange their path information with their direct neighbors periodically or when there is a change. They use the BGP algorithm to calculate the best paths. Examples are BGP, IDRP, etc.

Congestion control algorithms in computer networks

- Congestion control algorithms are mechanisms that control the entry of data packets into the network, enabling a better use of a shared network infrastructure and avoiding congestive collapse.
- Congestive collapse is a situation where the network performance degrades drastically due to excessive traffic and congestion.
- Congestion control algorithms can be broadly classified into two categories: open loop and closed loop.
- Open loop congestion control policies are applied to prevent congestion before it happens. They involve designing the network and choosing the appropriate protocols and parameters to avoid congestion.
- Closed loop congestion control policies are applied to detect and mitigate congestion after it happens. They involve monitoring the network state and adjusting the transmission rate or window size of the senders based

on feedback signals such as packet losses and delays.

- Some examples of open loop congestion control techniques are admission control, traffic shaping, and resource reservation.
- Some examples of closed loop congestion control techniques are congestion avoidance, congestion recovery, and congestion pricing.
- Congestion avoidance algorithms are implemented at the TCP layer as the mechanism to avoid congestive collapse in a network. They use additive increase/multiplicative decrease (AIMD) scheme, along with other schemes such as slow start and congestion window (CWND), to achieve congestion avoidance.
- Congestion recovery algorithms are implemented at the TCP layer as the mechanism to recover from packet losses due to congestion. They use fast retransmit, fast recovery, and selective acknowledgment (SACK) schemes to reduce the recovery time and improve the throughput.
- Congestion pricing algorithms are implemented at the network layer as the mechanism to allocate network resources based on the demand and supply. They use economic principles such as marginal cost pricing, smart market, and progressive second price auction to charge users for using the network and incentivize them to reduce their demand during congestion.

IPv6 in China

- IPv6 is the latest version of the Internet Protocol, which provides virtually unlimited IP addresses for electronic devices to communicate online.
- China is one of the leading countries in IPv6 deployment and application, with over 400 million active IPv6 users as of January 2023.
- China has set a goal of running a single-stack IPv6 network by 2030, which means that all devices and applications will use IPv6 only and not rely on IPv4.
- China has also issued a notice to accelerate the large-scale deployment and application of IPv6, which includes measures such as:
  - Promoting the upgrade of network infrastructure, such as routers, switches, firewalls, and domain name servers, to support IPv6.
  - Encouraging the development of IPv6-based applications and services, such as e-commerce, online education, cloud computing, and Internet of Things.
  - Enhancing the security and stability of IPv6 networks, such as preventing cyberattacks, data breaches, and privacy violations.
  - Strengthening the international cooperation and standardization of IPv6, such as participating in global forums, working groups, and testing activities .
- China's motivation for adopting IPv6 is mainly driven by the following factors:
  - The shortage of IPv4 addresses, which limits the growth of the internet and the innovation of new technologies.
  - The need to improve the quality and efficiency of internet services,

such as reducing latency, increasing bandwidth, and enabling end-to-end connectivity.
  - The ambition to become a global leader in the digital economy, such as enhancing the competitiveness, influence, and sovereignty of China in the cyberspace .

Unit 4 - Transport Layer in Computer Networks

- The transport layer is the fourth layer of the OSI model and the third layer of the TCP/IP model. It provides end-to-end communication services between applications running on different hosts in a network.
- The transport layer is responsible for the following functions:
  - Multiplexing and demultiplexing: The transport layer can use port numbers to identify different applications or processes on the same host and deliver data to the correct destination.
  - Segmentation and reassembly: The transport layer can divide a large message into smaller segments that fit into the network layer packets and reassemble them at the destination.
  - Connection-oriented and connectionless communication: The transport layer can support both reliable and unreliable data transfer modes. Connection-oriented communication involves establishing, maintaining, and terminating a logical connection between the sender and the receiver. Connectionless communication does not require any connection setup or teardown and sends data as independent datagrams.
  - Flow control: The transport layer can regulate the rate of data transmission between the sender and the receiver to avoid congestion and buffer overflow.
  - Error control: The transport layer can detect and correct errors in the data transmission using techniques such as checksums, acknowledgments, and retransmissions.
  - Congestion control: The transport layer can adapt to the varying network conditions and adjust the data transmission rate to avoid or reduce congestion in the network.
- The transport layer protocols can be classified into two categories: TCP and UDP.
  - TCP (Transmission Control Protocol) is a connection-oriented, reliable, and full-duplex protocol that provides reliable byte-stream service. It uses a three-way handshake to establish a connection, a sliding window mechanism to provide flow and error control, and a four-way handshake to terminate a connection. TCP also implements congestion control algorithms such as slow start, congestion avoidance, fast retransmit, and fast recovery.
  - UDP (User Datagram Protocol) is a connectionless, unreliable, and simplex protocol that provides datagram service. It does not guarantee delivery, order, or error-free transmission of data. It adds only a

minimal header to the data and sends it as a single datagram. UDP is suitable for applications that require speed, efficiency, or real-time communication, such as voice over IP, video streaming, or online gaming.

Process-to-process delivery in transport layer

- The transport layer is responsible for delivering data from one process to another process on different hosts across a network. This is called process-to-process delivery .
- A process is an application program that uses the services of the transport layer. For example, a web browser or an email client are processes that communicate with a web server or an email server respectively.
- The transport layer uses port numbers to identify the source and destination processes in a packet. A port number is a 16-bit integer that is added to the packet header by the transport layer .
- The transport layer also multiplexes and demultiplexes data from different processes. Multiplexing is the process of combining data from multiple processes into a single packet. Demultiplexing is the process of separating data from a single packet into multiple processes .
- The transport layer provides two types of services: connection-oriented and connectionless. Connection-oriented service establishes a logical connection between the source and destination processes before data transfer. Connectionless service does not require a connection and sends data as independent packets .
- The transport layer also performs error control, flow control, and congestion control. Error control is the process of detecting and correcting errors in the data packets. Flow control is the process of regulating the rate of data transmission between the source and destination processes. Congestion control is the process of avoiding or reducing network congestion by adjusting the data rate or dropping packets .

Transport layer protocols

- Transport layer protocols are responsible for providing end-to-end communication services for applications over a network.
- Transport layer protocols lie between the application layer and the network layer in the protocol stack, and they use the services of the lower layers to send and receive data packets.
- Transport layer protocols can be classified into two types: connection-oriented and connectionless.
- Connection-oriented protocols establish a logical connection between the sender and the receiver before exchanging data, and they ensure reliable and ordered delivery of data. An example of a connection-oriented protocol is the Transmission Control Protocol (TCP).
- Connectionless protocols do not require a prior connection between the

sender and the receiver, and they do not guarantee reliable or ordered delivery of data. An example of a connectionless protocol is the User Datagram Protocol (UDP).

- Transport layer protocols also provide other functions, such as flow control, congestion control, error detection, and multiplexing.
- Flow control is the mechanism that regulates the amount of data that the sender can transmit to the receiver, based on the receiver's buffer capacity and processing speed.
- Congestion control is the mechanism that prevents the network from being overloaded with too many packets, by adjusting the sending rate of the sender according to the network conditions.
- Error detection is the mechanism that detects and corrects errors in the data packets, such as lost, duplicated, corrupted, or out-of-order packets.
- Multiplexing is the mechanism that allows multiple applications to share the same transport layer protocol and the same network connection, by using port numbers to identify different processes.

UDP Transport layer protocol

- UDP stands for User Datagram Protocol and it is a transport layer protocol that is part of the Internet Protocol suite, also known as UDP/IP suite.
- UDP is a simple, unreliable and connectionless protocol that does not require a prior connection establishment or acknowledgment of data delivery .
- UDP provides a minimal amount of communication mechanism, such as adding transport-level addresses (ports), data lengths and error checking (checksums) to the data packets .
- UDP packets are called user datagrams and they have a 16-byte header that consists of four fields: source port, destination port, length and checksum .
- UDP is suitable for applications that need fast, efficient and low-overhead transmission of small amounts of data, such as real-time audio and video streaming, online gaming, DNS queries and VoIP calls .
- UDP does not provide any guarantees for message delivery, order, duplication or congestion control, and it leaves these functions to the upper layer protocols or the applications .
- UDP is one of the oldest transport layer protocols, and it is defined and documented in RFC 768.
- UDP is not the only transport layer protocol, and some other examples are TCP, DCCP and SCTP.

TCP Transport layer protocol

- TCP stands for Transmission Control Protocol. It is a transport layer protocol that facilitates the transmission of packets from source to destination.

- TCP is a connection-oriented protocol that means it establishes the connection prior to the communication that occurs between the computing devices in a network.
- TCP is a reliable protocol as it follows the flow and error control mechanism. It also supports the acknowledgment mechanism, which checks the state and sound arrival of the data.
- TCP has three main steps to perform the data transmission: establish connection, send packets of data, and close the connection.
- TCP uses a four-way handshake to establish the connection between the sender and the receiver. The sender sends a SYN (synchronize) segment, the receiver replies with a SYN-ACK (synchronize-acknowledge) segment, the sender responds with an ACK (acknowledge) segment, and the connection is established.
- TCP uses a sliding window protocol to send packets of data. The sender can send multiple packets without waiting for the acknowledgment of each one, but it has a limit on how many packets it can send before receiving an acknowledgment. The receiver can send an acknowledgment for multiple packets at once, and also request the sender to retransmit any lost or corrupted packets.
- TCP uses a three-way handshake to close the connection between the sender and the receiver. The sender sends a FIN (finish) segment, the receiver replies with an ACK segment, the receiver sends a FIN segment, and the sender replies with an ACK segment. The connection is then terminated.
- TCP is a standard that defines how to establish and maintain a network conversation through which application programs can exchange data.
- TCP is used by many applications that require reliable and ordered delivery of data, such as web browsers, email clients, file transfer programs, etc.

Multiplexing in transport layer

- Multiplexing is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver.
- Multiplexing in transport layer means extending the host-to-host delivery service provided by the network layer to a process-to-process delivery service for applications running on the hosts.
- A multiplexing-demultiplexing service is needed for all computer networks, because a host can run multiple processes that need to communicate with processes on other hosts.
- Multiplexing in transport layer requires that sockets have unique identifiers, and each segment have special fields that indicate the sockets to which the segment is to be delivered.
- There are two types of multiplexing in transport layer: connectionless

multiplexing and connection-oriented multiplexing.

- Connectionless multiplexing uses the UDP protocol, which does not establish a connection before sending data. The sender adds the destination port number to the segment header, and the receiver uses the port number to deliver the segment to the correct socket.
- Connection-oriented multiplexing uses the TCP protocol, which establishes a connection before sending data. The sender and the receiver exchange segments with source and destination port numbers, as well as sequence and acknowledgment numbers, to identify and order the segments belonging to the same connection.
- Multiplexing in transport layer allows multiple applications to share the network resources and communicate with each other efficiently and reliably.

Connection management in transport layer

- The transport layer is responsible for providing reliable and efficient communication between two end systems (hosts) over a network.
- Connection management is the process of establishing, maintaining, and terminating a logical connection between two end systems.
- Connection management involves three phases: connection establishment, data transfer, and connection termination.
- Connection establishment is the phase where the two end systems agree on the parameters and rules for the data transfer, such as the sequence numbers, window sizes, and acknowledgment schemes.
- Data transfer is the phase where the actual data is exchanged between the two end systems, using the agreed parameters and rules. The transport layer ensures that the data is delivered reliably and in order, and handles any errors, losses, or delays that may occur in the network.
- Connection termination is the phase where the two end systems signal each other that they have no more data to send, and gracefully close the connection. The transport layer ensures that all the data has been successfully delivered and acknowledged, and releases the resources allocated for the connection.
- Connection management can be either connection-oriented or connectionless, depending on the transport layer protocol used.
- Connection-oriented protocols, such as TCP, require a connection to be established before any data can be transferred, and maintain the connection until it is explicitly terminated. Connection-oriented protocols provide reliable and ordered delivery of data, but incur more overhead and delay due to the connection setup and teardown.
- Connectionless protocols, such as UDP, do not require a connection to be established or terminated, and allow data to be transferred at any time without any prior agreement. Connectionless protocols provide fast and efficient delivery of data, but do not guarantee reliability or order, and may result in data loss, duplication, or reordering.

Flow control in transport layer

- Flow control is the mechanism that regulates the rate of data transmission between two nodes in a network.
- Flow control is needed in transport layer because it prevents data loss due to buffer overflow or underflow, and improves the efficiency and reliability of data transmission.
- Flow control in transport layer is different from flow control in data link layer, because it operates end-to-end, not just across a single link.
- Flow control in transport layer can be implemented by using feedback-based or rate-based techniques.
- Feedback-based flow control uses acknowledgments and windowing to adjust the sender's transmission rate according to the receiver's buffer capacity and network congestion.
- Rate-based flow control uses a predefined transmission rate that is agreed upon by both sender and receiver, and does not depend on feedback.
- Some examples of transport layer protocols that use flow control are TCP, UDP, and SCTP.

etransmission in transport layer

- The transport layer is the fourth layer in the OSI model and the third layer in the TCP/IP model. It provides end-to-end communication services for applications, such as reliable data transfer, flow control, congestion control, and error detection and correction.
- The transport layer uses protocols such as TCP and UDP to segment the data from the application layer into smaller units called segments or datagrams, and to add headers that contain information such as source and destination port numbers, sequence numbers, and checksums.
- The transport layer is responsible for ensuring that the data is transmitted reliably and efficiently across the network. It uses various mechanisms to achieve this, such as:
  - Acknowledgements: The receiver sends back a message to the sender to confirm the receipt of a segment or datagram. This helps the sender to keep track of the data that has been delivered and to avoid sending duplicate data.
  - Retransmission: The sender re-sends a segment or datagram that has not been acknowledged by the receiver within a certain time interval. This helps to recover from data loss due to network errors or congestion .
  - Flow control: The sender adjusts the rate of data transmission according to the feedback from the receiver and the network conditions. This helps to prevent the sender from overwhelming the receiver or the network with too much data.
  - Congestion control: The sender reduces the rate of data transmission when the network is congested, and increases it when the network is

less congested. This helps to avoid or alleviate network congestion and improve the overall performance of the network.

- The transport layer enables different applications to communicate with each other using port numbers. A port number is a 16-bit number that identifies a specific application or service on a host. The transport layer header contains the source and destination port numbers, which are used by the network layer to deliver the data to the correct application or service.

Window management in transport layer

- The transport layer is the fourth layer of the OSI model and provides end-to-end communication services for applications.
- The transport layer is responsible for flow control, which is the regulation of the rate and order of data transmission between two hosts.
- Flow control is necessary to prevent congestion, loss, and reordering of packets in the network.
- One of the techniques used by the transport layer for flow control is the sliding window technique, which is implemented by the Transmission Control Protocol (TCP).
- The sliding window technique involves the use of a window, which is a range of sequence numbers that indicate the packets that can be sent or received by a host at any given time.
- The window size can vary depending on the buffer availability, network conditions, and feedback from the receiver.
- The sender maintains a send window, which indicates the packets that can be sent without waiting for an acknowledgment from the receiver.
- The receiver maintains a receive window, which indicates the packets that can be received and buffered without overflowing the buffer.
- The sender and receiver exchange window information using special control packets called acknowledgments (ACKs).
- The sender adjusts its send window based on the receive window advertised by the receiver in the ACKs.
- The receiver adjusts its receive window based on the packets received and buffered.
- The sliding window technique allows the sender and receiver to dynamically adapt to the network conditions and achieve optimal throughput and reliability.

TCP Congestion control in transport layer

- TCP congestion control is a mechanism that aims to regulate the amount of data sent by a sender to avoid overwhelming the network or the receiver.
- TCP congestion control is based on the concept of congestion window (cwnd), which is the maximum number of bytes that a sender can have in flight (unacknowledged) at any time.

- TCP congestion control consists of four phases: slow start, congestion avoidance, fast retransmit, and fast recovery.
- Slow start: The sender starts with a small cwnd (usually one segment) and increases it by one segment for every ACK received, until it reaches a threshold (ssthresh) or a loss occurs.
- Congestion avoidance: The sender increases the cwnd by a fraction of a segment for every ACK received, to probe the network capacity gradually.
- Fast retransmit: The sender detects a loss by receiving three duplicate ACKs for the same segment, and retransmits the lost segment without waiting for a timeout.
- Fast recovery: The sender reduces the cwnd by half and enters the congestion avoidance phase, to recover from the loss quickly.

Quality of service in transport layer

- Quality of service (QoS) is the use of mechanisms or technologies that work on a network to control traffic and ensure the performance of critical applications with limited network capacity.
- The transport layer is responsible for enhancing the QoS provided by the network layer by offering reliable and efficient end-to-end data delivery.
- The transport layer can provide QoS in terms of:
  - Throughput: the rate of data transfer between the sender and the receiver.
  - Delay: the time it takes for a packet to travel from the sender to the receiver.
  - Jitter: the variation in delay of packets.
  - Reliability: the probability of successful delivery of packets.
  - Security: the protection of data from unauthorized access or modification.
- The transport layer can use different protocols and techniques to achieve QoS, such as:
  - Transport Control Protocol (TCP): a connection-oriented protocol that provides reliable, ordered, and error-free data delivery. TCP uses mechanisms such as flow control, congestion control, and error recovery to ensure QoS.
  - User Datagram Protocol (UDP): a connectionless protocol that provides fast and efficient data delivery. UDP does not guarantee reliability, ordering, or error-free delivery, but it has lower overhead and latency than TCP. UDP is suitable for real-time applications that can tolerate some packet loss.
  - Stream Control Transmission Protocol (SCTP): a connection-oriented protocol that provides reliable, unordered, and error-free data delivery. SCTP supports multiple streams of data within a single connection, which can reduce jitter and improve QoS for multimedia applications.
  - Transport Layer Security (TLS): a protocol that provides security

and encryption for data transmitted over TCP. TLS can protect data from eavesdropping, tampering, and impersonation attacks.
  - Quality of Service Transport Protocol (QSTP): a protocol that provides QoS for wireless sensor networks. QSTP uses a cross-layer approach that integrates the transport layer and the network layer to achieve QoS in terms of throughput, delay, reliability, and energy efficiency.

Unit 5 - Application Layer in Computer Networks

- The application layer is the topmost layer of the TCP/IP model or the OSI model that provides the interface between the applications and the network.
- The application layer defines the protocols and services that enable communication and data exchange between different applications running on different hosts or devices.
- The application layer protocols are specific to the type of application and data being transmitted, such as HTTP for web browsing, SMTP for email, FTP for file transfer, DNS for name resolution, etc.
- The application layer protocols use the services of the transport layer protocols, such as TCP or UDP, to establish reliable or unreliable connections and to segment, reassemble, and order the data packets.
- The application layer protocols also use the services of the lower layers, such as the network layer, the data link layer, and the physical layer, to route, encode, and transmit the data packets across the network.
- The application layer is responsible for some of the following functions:
  - Application identification: The application layer identifies the type and format of the data being transmitted and the appropriate protocol to use for communication.
  - Data encoding and compression: The application layer encodes the data into a standard format, such as ASCII or Unicode, and compresses the data to reduce the bandwidth and storage requirements.
  - Data encryption and decryption: The application layer encrypts the data to ensure confidentiality and integrity, and decrypts the data at the destination using the appropriate keys and algorithms.
  - Data segmentation and reassembly: The application layer segments the data into smaller units, such as messages or datagrams, and adds headers and trailers to each unit. The application layer also reassembles the data units at the destination and removes the headers and trailers.
  - Data presentation and interpretation: The application layer presents the data to the user in a meaningful and understandable way, such as text, images, audio, or video. The application layer also interprets the data and performs the appropriate actions, such as displaying a web page, sending an email, or playing a song.
  - Error detection and correction: The application layer detects and cor-

rects any errors that may occur during the data transmission, such as lost, duplicated, or corrupted packets. The application layer may use techniques such as checksums, acknowledgments, or retransmissions to ensure the accuracy and completeness of the data.
  - Session management: The application layer manages the sessions between the applications and the network, such as establishing, maintaining, and terminating the connections. The application layer may use techniques such as cookies, tokens, or timestamps to identify and authenticate the users and the applications.
  - Service discovery and advertisement: The application layer discovers and advertises the available services and resources on the network, such as printers, servers, or files. The application layer may use techniques such as multicast, broadcast, or directory services to locate and access the services and resources.

Domain Name System

- The Domain Name System (DNS) is a hierarchical and decentralized system that translates human-readable domain names (such as www.example.com) into numerical IP addresses (such as 192.0.2.1) that identify the devices and services on the Internet.
- The DNS consists of four main components: domain names, name servers, resolvers, and records.
- Domain names are the alphanumeric labels that identify a domain or a subdomain on the Internet. For example, example.com is a domain name, and www.example.com is a subdomain of example.com. Domain names are organized into a tree-like structure, with the root domain at the top, followed by top-level domains (TLDs), second-level domains (SLDs), and so on. Each level is separated by a dot (.).
- Name servers are the servers that store and provide the DNS records for a domain or a subdomain. Name servers are authoritative for the domains or subdomains they serve, meaning they have the final and definitive answer for any DNS query. Name servers are also responsible for delegating authority to other name servers for lower-level domains or subdomains. For example, the name servers for example.com can delegate authority to the name servers for www.example.com.
- Resolvers are the clients that query the name servers for DNS records. Resolvers can be either recursive or iterative. Recursive resolvers send queries to the root name servers, and then follow the referrals from the root name servers to the TLD name servers, and then to the SLD name servers, and so on, until they reach the authoritative name server for the requested domain or subdomain. Iterative resolvers send queries to a specified name server, and expect either an authoritative answer or a referral to another name server. Iterative resolvers can use caching to store and reuse the DNS records they receive from the name servers.
- Records are the data elements that map a domain name to an IP address

or other information. Records have different types, such as A, AAAA, CNAME, MX, NS, SOA, TXT, etc. Each record type has a specific format and purpose. For example, an A record maps a domain name to an IPv4 address, a CNAME record maps a domain name to another domain name, an MX record maps a domain name to a mail server, etc. Records also have a time-to-live (TTL) value, which specifies how long the record can be cached by the resolvers.

World Wide Web

- The World Wide Web (WWW) is a global information system that uses the Internet to link documents and resources across different devices and locations.
- The WWW was invented by Tim Berners-Lee in 1989 at CERN, the European Organization for Nuclear Research.
- The WWW consists of three main components: Uniform Resource Identifiers (URIs), Hypertext Transfer Protocol (HTTP), and Hypertext Markup Language (HTML).
- URIs are strings of characters that identify and locate resources on the web, such as web pages, images, videos, etc. URIs usually start with a scheme name, such as http, https, ftp, mailto, etc., followed by a colon and a specific syntax for the scheme.
- HTTP is the protocol that defines how web browsers and web servers communicate and exchange data. HTTP uses a request-response model, where a client (browser) sends a request to a server and the server responds with a status code and optionally some data.
- HTML is the language that defines the structure and content of web pages. HTML uses tags, which are enclosed in angle brackets, to mark up different elements, such as headings, paragraphs, links, images, etc. HTML also supports Cascading Style Sheets (CSS) and JavaScript, which are used to style and add interactivity to web pages.

Hyper Text Transfer Protocol

- Hyper Text Transfer Protocol (HTTP) is a protocol that defines how web browsers and web servers communicate and exchange data over the internet.

- HTTP is based on a request-response model, where a client (such as a browser) sends a request to a server (such as a web server) and the server responds with a status code, headers, and optionally a body.

- HTTP requests and responses can contain different types of data, such as text, images, audio, video, etc. The data type is specified by the Content-Type header.

- HTTP uses Uniform Resource Locators (URLs) to identify the resources that the client wants to access. A URL consists of a scheme (such as http or https), a host name (such as www.example.com), a port number (optional), a path (such as /index.html), and a query string (optional).

- HTTP supports different methods (also called verbs) to perform different actions on the resources. The most common methods are GET, POST, PUT, DELETE, HEAD, and OPTIONS.
- GET method is used to retrieve a resource from the server. The request can include query parameters in the URL to specify the criteria for the resource.
- POST method is used to send data to the server, such as form data, files, etc. The request can include a body that contains the data to be sent.
- PUT method is used to update or create a resource on the server. The request can include a body that contains the data to be updated or created.
- DELETE method is used to delete a resource from the server. The request can include query parameters in the URL to specify the criteria for the resource.
- HEAD method is used to get the headers of a resource from the server, without the body. This can be useful to check the metadata of a resource, such as its size, type, modification date, etc.
- OPTIONS method is used to get the allowed methods for a resource from the server. This can be useful to check the capabilities of a resource, such as what methods it supports, what headers it accepts, etc.
- HTTP uses status codes to indicate the outcome of a request. The status codes are divided into five categories: 1xx (informational), 2xx (success), 3xx (redirection), 4xx (client error), and 5xx (server error).
- Some of the common status codes are:
  - 200 OK: The request was successful and the response contains the requested resource.
  - 301 Moved Permanently: The requested resource has been moved to a new URL and the client should use the new URL for future requests.
  - 302 Found: The requested resource has been found at a different URL and the client should use the new URL for this request only.
  - 304 Not Modified: The requested resource has not been modified since the last request and the client can use the cached version of the resource.
  - 400 Bad Request: The request was malformed or invalid and the server could not process it.
  - 401 Unauthorized: The request requires authentication and the client did not provide valid credentials.
  - 403 Forbidden: The request was valid but the server refused to fulfill it due to authorization or permission issues.

- 404 Not Found: The requested resource was not found on the server and the server does not know where to find it.
- 500 Internal Server Error: The server encountered an unexpected error while processing the request and could not complete it.
- 503 Service Unavailable: The server is temporarily unable to handle the request due to overload or maintenance and the client should try again later.

Electronic mail in application layer

- Electronic mail (or email) is an application layer service that allows users to exchange messages and information over the internet.
- Email is one of the most popular and widely used services of the internet.
- Email has three major components: user agents, mail servers, and protocols.
- User agents are the software programs that users use to read, compose, and organize email messages. Examples of user agents are Outlook, Gmail, and Thunderbird.
- Mail servers are the servers that store and forward email messages. Each user has a mailbox on a mail server that belongs to their domain. For example, a user with the email address alice@example.com has a mailbox on the mail server example.com.
- Protocols are the rules and standards that govern the communication between user agents and mail servers, and between mail servers themselves. The main protocols used for email are:
  - Simple Mail Transfer Protocol (SMTP): SMTP is used to send email messages from a user agent to a mail server, or from one mail server to another. SMTP uses port 25 and operates over TCP. SMTP follows a push model, where the sender initiates the transfer of a message to the receiver.
  - Post Office Protocol (POP): POP is used to retrieve email messages from a mail server to a user agent. POP uses port 110 and operates over TCP. POP follows a pull model, where the receiver requests the transfer of a message from the sender. POP usually deletes the messages from the server after downloading them to the user agent.
  - Internet Message Access Protocol (IMAP): IMAP is similar to POP, but it allows more flexibility and functionality. IMAP uses port 143 and operates over TCP. IMAP allows the user to access email messages without downloading them, and to synchronize multiple user agents with the same mailbox. IMAP also supports email folders, flags, and searches. IMAP does not delete the messages from the server unless the user explicitly does so.
  - Multipurpose Internet Mail Extensions (MIME): MIME is not a protocol, but a standard that defines the format and structure of email messages. MIME allows email messages to contain different types of content, such as text, images, audio, video, and attachments.

MIME also supports different character sets, languages, and encodings. MIME defines a set of headers and boundaries that separate the different parts of a message.

File Transfer Protocol in Application Layer

- File Transfer Protocol (FTP) is an application layer protocol that is used to transfer files between local and remote file systems over the Internet .
- FTP runs on top of TCP, which provides reliable and ordered delivery of data packets .
- FTP uses two TCP connections in parallel to transfer a file: a control connection and a data connection .
- The control connection is used to exchange commands and responses between the FTP client and the FTP server .
- The data connection is used to transfer the actual file data between the FTP client and the FTP server .
- The control connection remains open throughout the FTP session, while the data connection is opened and closed for each file transfer .
- FTP supports both text and binary files, and can handle different types of file systems and end-of-line characters .
- FTP requires the user to authenticate with a username and password before accessing the files on the server .
- FTP uses a plaintext (unencrypted) sign-in process, which makes it vulnerable to eavesdropping and spoofing attacks .
- FTP can be secured by using encryption protocols such as SSL/TLS or SSH .

Remote login in application layer

- Remote login is a process that allows an authorized user to access and use the services of another computer over a network, as if the user were physically present at the remote computer.
- Remote login is an example of an application layer service, which is the highest layer in the network protocol stack. The application layer provides the interface and functionality for various network applications, such as email, web browsing, file transfer, etc.
- Remote login can be implemented using different protocols and methods, such as Telnet, SSH, Rlogin, etc. Each protocol has its own advantages and disadvantages, such as security, performance, compatibility, etc.
- Remote login can be useful for various purposes, such as:
  - Remote administration: A user can manage and configure a remote computer or server without being physically present.
  - Remote access: A user can access and use the resources and files of a remote computer or server, such as databases, documents, applications, etc.
  - Remote collaboration: A user can share and work on the same files

  and applications with other users who are logged in to the same remote computer or server.
  - Remote education: A user can attend and participate in online classes and courses that are hosted on a remote computer or server.
  - Remote entertainment: A user can play games and watch videos that are streamed from a remote computer or server.

Network management in application layer

- The application layer is the topmost layer in the Open System Interconnection (OSI) model, which defines how different devices communicate over a network.
- The application layer provides several ways for manipulating the data (information) which actually enables any type of user to access network with ease.
- The application layer handles network services, such as file and printing, name resolution, and redirector services. Name resolution is the process of mapping an IP address to a human-readable name.
- The application layer is where users interact with the network, download information and send data. The application layer also provides protocols for different types of applications, such as email, web browsing, file transfer, etc.
- Network management is the process of configuring, monitoring, and managing the performance of a network. Network management also ensures that network resources, such as hardware, storage, memory, bandwidth, data and processing power, are made readily accessible to users and services as efficiently and securely as possible.
- Network management can be performed at different layers of the OSI model, but the application layer is especially important because it is the interface between the network and the end users. Network management at the application layer involves tasks such as:
  - Providing user authentication and authorization
  - Ensuring data integrity and confidentiality
  - Monitoring and optimizing network performance and availability
  - Troubleshooting and resolving network issues
  - Implementing network policies and security measures
  - Managing network services and applications
- Network management at the application layer requires a platform that can support various protocols, tools, and processes. Some examples of network management platforms are Cisco Network Services Orchestrator, IBM Cloud Pak for Network Automation, and SolarWinds Network Performance Monitor.

Data compression in application layer

- Data compression is the function of presentation layer in OSI reference model.
- The presentation layer structures data that is passed down from the application layer into a format suitable for network transmission.
- Data compression allows to reduce the number of bits that needs to be transmitted on the network, which can improve the efficiency and performance of data communication .
- Data compression can be performed by various algorithms, such as Huffman coding, run-length encoding, Lempel-Ziv-Welch (LZW) algorithm, etc.
- Data compression can be applied to different types of data, such as text, images, audio, video, etc.
- Data compression can be lossless or lossy, depending on whether the original data can be recovered exactly or not after decompression.
- Data compression can be symmetric or asymmetric, depending on whether the same or different algorithms are used for compression and decompression.
- Data compression can be adaptive or static, depending on whether the compression algorithm adapts to the characteristics of the data or not.
- Data compression can be performed at the application layer by some protocols, such as File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP), Domain Name System (DNS), etc .
- Data compression at the application layer can provide more flexibility and control over the compression process, but it can also increase the complexity and overhead of the application .

Cryptography in application layer

- Cryptography is the process of converting plain text into cipher text, which is unintelligible and vice-versa.
- Cryptography provides secure communication in the presence of adversaries by using encryption and decryption algorithms and keys.
- Application layer encryption is a data-security solution that encrypts nearly any type of data passing through an application .
- When encryption occurs at the application layer, data is encrypted across multiple (including disk, file, and database) layers .
- This application layer encryption approach increases security by reducing the number of potential attack vectors.
- Application layer encryption also gives developers more control over what gets encrypted and who gets the keys for decryption.
- End-to-end encryption is an increasingly popular type of application layer encryption that lets organizations enforce access control using key management as well as policy.
- In some cases, the users themselves may be the only parties with the keys

for decryption.

- Application layer encryption can be used in native and browser applications for various purposes, such as protecting sensitive data, ensuring data integrity, verifying data origin, and authenticating users .

Basic concepts of cryptography in application layer

Cryptography is the science of securing communications from unauthorized parties. It involves the use of mathematical techniques to transform plain text into cipher text, which is unintelligible, and vice versa. Cryptography can provide confidentiality, integrity and authenticity to the data transmitted or stored in applications.

Some of the basic concepts of cryptography in application layer are:

- **Symmetric key cryptography**: This is a type of cryptography where the same key is used for both encryption and decryption. The key must be shared securely between the communicating parties. Symmetric key cryptography is fast and efficient, but it suffers from the key distribution problem, which is how to securely exchange the key without compromising it. Examples of symmetric key algorithms are AES, DES, RC4, etc.

- **Asymmetric key cryptography**: This is a type of cryptography where a pair of keys is used for encryption and decryption. One key is called the public key, which can be freely distributed, and the other is called the private key, which must be kept secret. The public key can be used to encrypt a message, which can only be decrypted by the corresponding private key. The private key can also be used to sign a message, which can be verified by the public key. Asymmetric key cryptography solves the key distribution problem of symmetric key cryptography, but it is slower and more complex. Examples of asymmetric key algorithms are RSA, ECC, DSA, etc.

- **Hash functions**: These are mathematical functions that map an arbitrary input to a fixed-length output, called the hash or digest. Hash functions are one-way, meaning that it is easy to compute the hash from the input, but hard to find the input from the hash. Hash functions can be used to provide integrity to the data, by verifying that the data has not been tampered with. Examples of hash functions are SHA, MD5, etc.

- **Digital signatures**: These are a way of using asymmetric key cryptography to provide authenticity and non-repudiation to the data. A digital signature is a value that is computed from the data and the signer's private key, and attached to the data. Anyone who has the signer's public key can verify the signature and the data. A digital signature proves that the data was signed by the owner of the private key, and that the data was not altered after signing. Examples of digital signature algorithms are RSA, DSA, ECDSA, etc.

- **Encryption modes**: These are ways of applying symmetric key cryptography to encrypt or decrypt a stream of data. Different encryption modes have different properties and trade-offs, such as security, performance, error propagation, etc. Examples of encryption modes are ECB, CBC, CTR, GCM, etc.
- **Cryptographic protocols**: These are rules and procedures that define how cryptography is used in applications to achieve specific goals, such as secure communication, authentication, key exchange, etc. Cryptographic protocols often involve multiple steps and parties, and use various cryptographic primitives and techniques. Examples of cryptographic protocols are SSL/TLS, SSH, PGP, etc.